

Hydraulic Fracturing's Dirty Secret

(Wastewater Treatment)

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Introduction

Objective

As urbanization, population, and energy production demand continues to increase, fresh water supply continues to rise in high demand. Fresh water is no longer an unlimited resource that we all formerly assumed it is. By 2040, the demand for fresh water is expected to surpass supply to about 40% [1]. The time to apply innovative ideas in treating wastewater is now, as our supply of fresh water continues to diminish rapidly. One of the largest contributors of wastewater on the industrial scale is from hydraulic fracturing. Hydraulic fracturing is the process of injecting large amounts of water mixed with sand and chemical additives at high pressure to create cracks in deep-rock formations. Fracking is used to retrieve natural gas and oil that is trapped within the formations. As the popularity of “fracking” increases in the United States, the amount of water that is used continues to escalate. About 2.4 billion gallons of wastewater pour out of oil and gas wells in the U.S. daily [2]. The aim of this report is to compare the different solutions in treating wastewater from fracking. Some of these solutions include deep-well injection, storage in surface ponds, and treating the wastewater for reuse in the fracking process. The primary focus will be creating a mobile solution that can travel to different fracking sites to treat the flowback wastewater onsite during the fracking process. Having a mobile solution to treat and recycle fracking wastewater will help reduce the need to purchase fresh water, as well as reduce the environmental risk that storage of wastewater poses. This solution will revolutionize fracking by enabling billions of gallons of water to be retained and reused, thereby becoming an essential wastewater treatment that is vital for the future.

Market Analysis

The incentive for Hydraulic Fracturing Wastewater Treatment does not arise just from the environmental concerns. Such a process also has economic benefits. The market for more sustainable options in industry have gained more momentum as sustainability increases in importance due to the correlation of well-being and environment conservation for those that inhabit Earth. As people better understand the harmful effects of contamination and pollution, more regulations at state and government levels are being established regarding treating and disposing harmful components in water. Industrial water use is second in the world's water use [3] and first in North America. The other top ways that the world utilizes water is agriculture and municipal. Therefore, the need to find economically feasible systems to treat wastewater are a high priority when it comes to the industrial water sector. If more water is taken from the accessible sources and sent back with contaminants the issues and repercussions are more costly than those that can be quantified due to severe biological impairments that surface from high harmful “Total Dissolved Solids” (TDS’) [4]. Hence, treating the flowback water from Hydraulic Fracking is one step towards addressing a growing industry in the United State with a solution that saves money and meets costly regulations. Plus, the ability to transform a typically stationary wastewater treatment to a mobile system enables the economic flexibility to profit even more in the long run. The graphic below displays the World’s Fresh Water Resource to understand the order of magnitude in which action needs to be taken to conserve potable water source that is only 2.5% of the total water on Earth. Also, the graphic proceeding it shows that contamination should be avoided since groundwater makes up 1.69% [5] which is over half of the total freshwater sources. Lakes make up 0.013% of the total, while rivers make up 0.0002% [5] and these are major sources

for the water used for Fracking; therefore, treating the flowback water will aid in reducing the need to take water from an already scarce source.

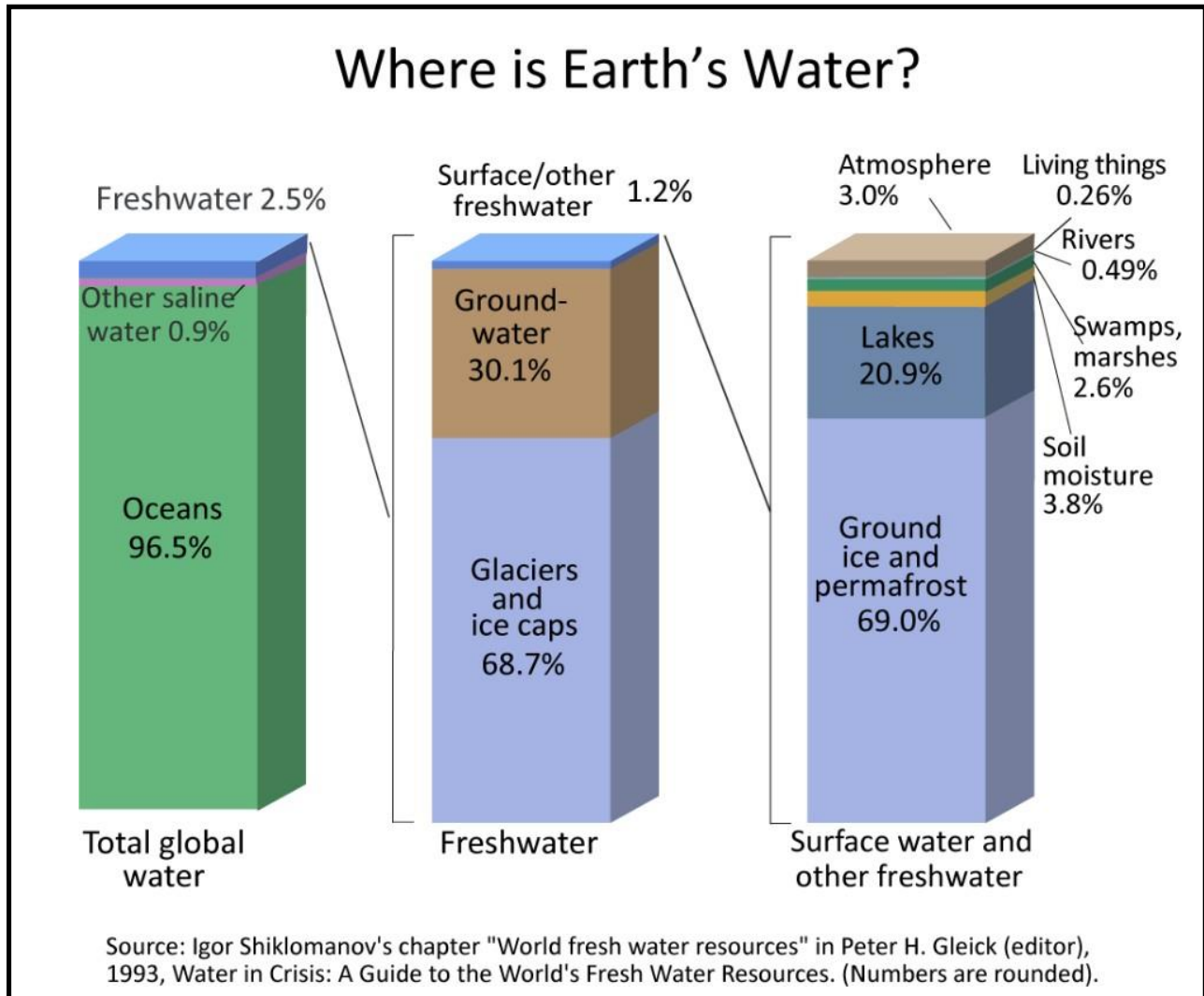


Figure 1: Earth's Water Sources [6]

Water source	Water volume, in cubic miles	Water volume, in cubic kilometers	Percent of freshwater	Percent of total water
Oceans, Seas, & Bays	321,000,000	1,338,000,000	--	96.54
Ice caps, Glaciers, & Permanent Snow	5,773,000	24,064,000	68.7	1.74
Groundwater	5,614,000	23,400,000	--	1.69
<i>Fresh</i>	<i>2,526,000</i>	<i>10,530,000</i>	<i>30.1</i>	<i>0.76</i>
<i>Saline</i>	<i>3,088,000</i>	<i>12,870,000</i>	--	<i>0.93</i>
Soil Moisture	3,959	16,500	0.05	0.001
Ground Ice & Permafrost	71,970	300,000	0.86	0.022
Lakes	42,320	176,400	--	0.013
<i>Fresh</i>	<i>21,830</i>	<i>91,000</i>	<i>0.26</i>	<i>0.007</i>
<i>Saline</i>	<i>20,490</i>	<i>85,400</i>	--	<i>0.006</i>
Atmosphere	3,095	12,900	0.04	0.001
Swamp Water	2,752	11,470	0.03	0.0008
Rivers	509	2,120	0.006	0.0002
Biological Water	269	1,120	0.003	0.0001

Source: Igor Shiklomanov's chapter "World fresh water resources" in Peter H. Gleick (editor), 1993, *Water in Crisis: A Guide to the World's Fresh Water Resources* (Oxford University Press, New York).

Figure 2: World Fresh Water Resources [5]

Value Chain

Analyzing the flow of water consumption, the graphic below displays how surface water and groundwater are both sources for Industrial use. Hydraulic Fracturing is included in the mining consumers in the graphic because it is associated with drilling to extract the oil or natural gas. The mining industry consumes 4,000 Mgal/day [7] and, at an average cost per gallon of water in 2020 being \$0.006, that means that each day about 24 million dollars are spent on water [8]. That cost does not include transporting the water to and from the site. When including \$0.11 per tonne per

mile [9] to transport the flowback water to treatment offsite it adds about \$4,619,422 per mile, estimated around 50 miles of distance that would be \$230,971,128 in just one day added to the \$24 million that makes it around \$255 million. And to transport the water to the Fracking site adds another cost of around \$200 per truck [10] and assuming each truck can hold 11,000 gallons [11] that would mean around 727 million dollars are added to the cost. The grand total of purchasing water, delivering it to the site, and then transporting the wastewater for treatment adds up to be 981 million dollars.

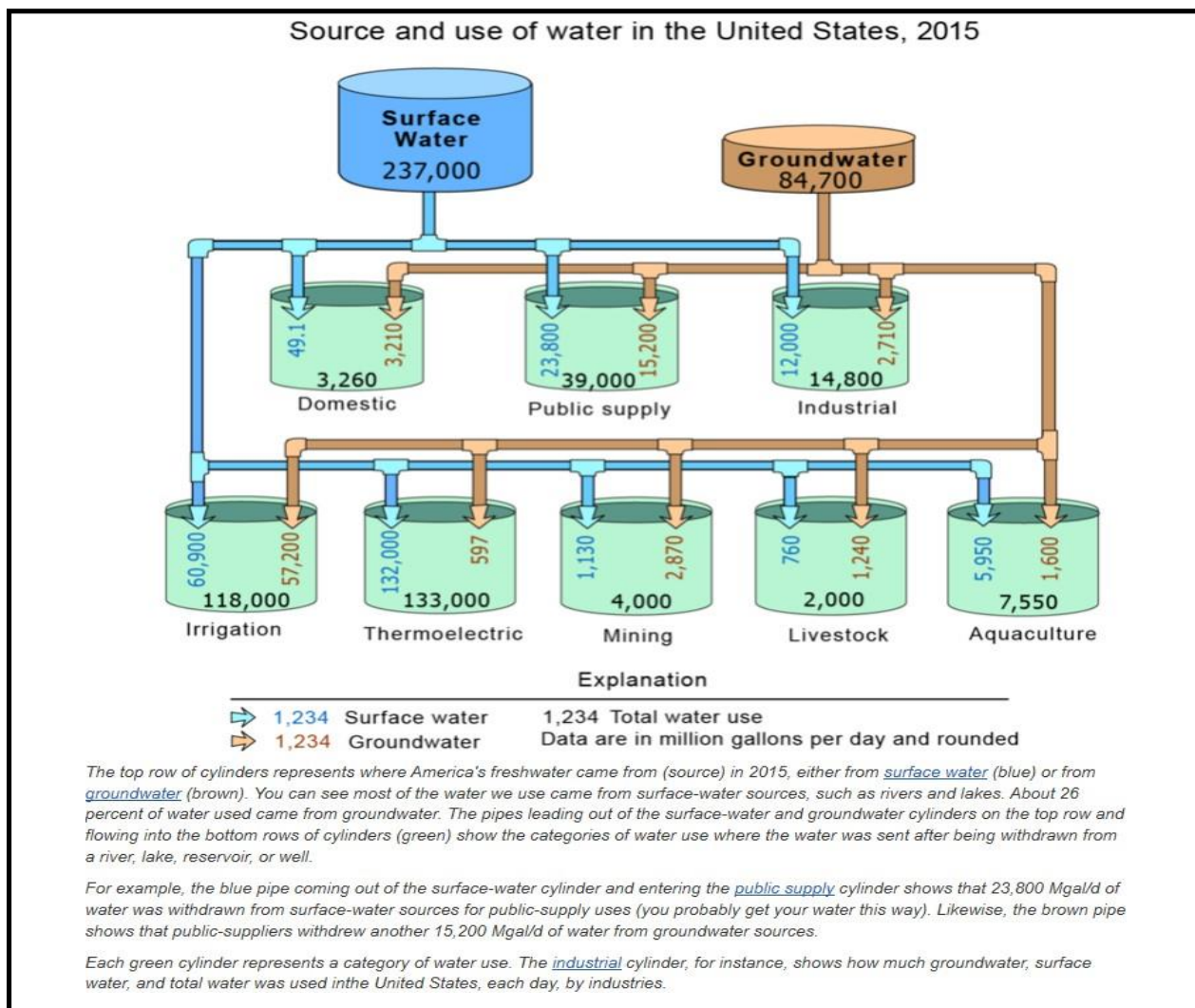


Figure 3:Source and Use of Water in the United States, 2015 [7]

The above calculation illustrates that, while the value of water initially seems minute, when billions of gallons are used and associated expenses are included, the cost of the water becomes significant and the source dwindles. Implementing an environmentally friendly, mobile system to treat flowback water from fracking will reduce the cost from obtaining, transporting, and treating the wastewater. This system would also allow for Hydraulic Fracturing companies to treat and produce water meeting the standards required to reuse the water for fracking. Figure 4 below displays the value chain for the oil and gas industry [12].

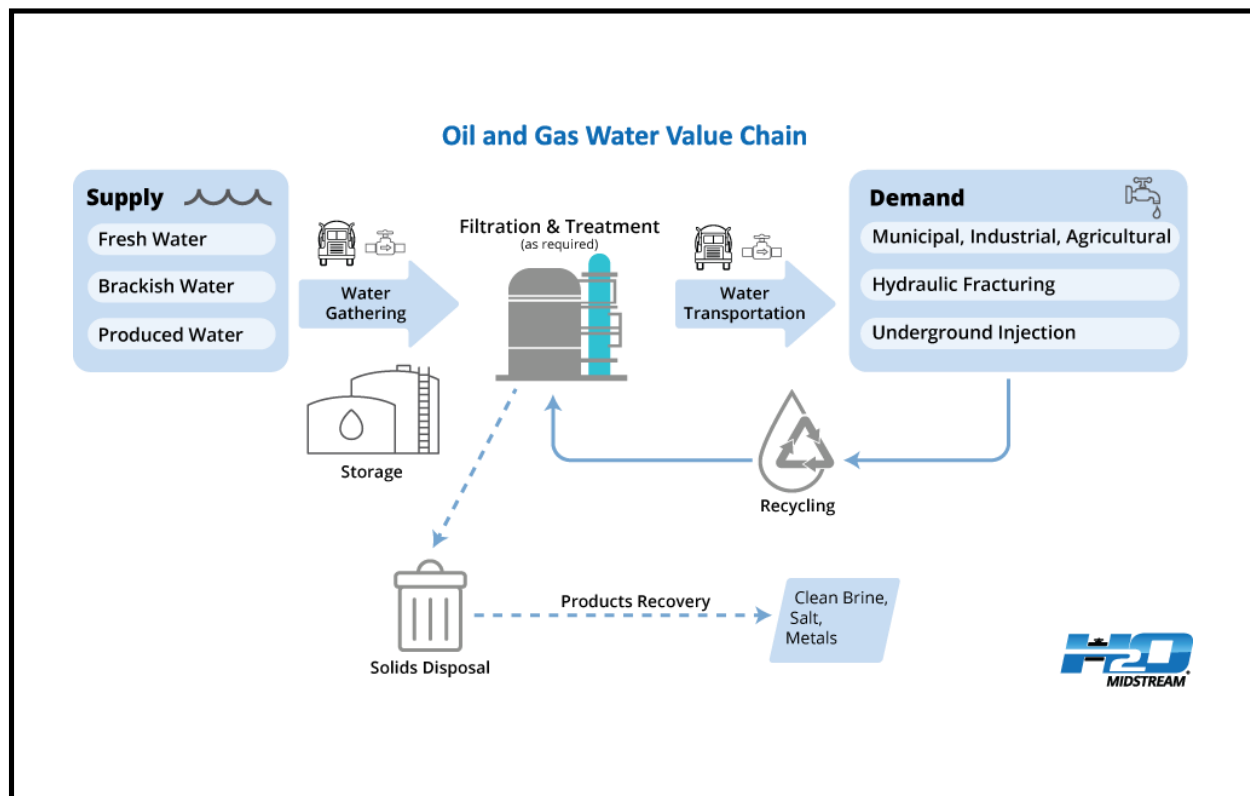


Figure 4: Value chain for the oil and gas industry [12]

Market Drivers

The main market driver for water treatment implementation is the “need for economical and sustainable oil field water management, which has led to the rise of midstream water management companies. Their services include treatment, storage and efficient transportation of water to oil fields, as well as re-use” [13]. The growth in Hydraulic Fracturing opens up a market to be innovative and proactive by providing a mobile system with improved overall treatment cost. The complete global water and wastewater market is summarized in the figure below, which shows that the 2019 forecast was \$743.76 billion total based on 2018 data. Of the total, \$368.16 billion was expenditures by Industry and Municipal, of which the industrial expenditure was \$179.51 billion. Assuming the growth from 2019 to 2020 will be the same as the growth from 2018 to 2019, the Industrial Market Expenditure Forecast for 2020 is \$193.15 billion. This includes chemicals, process control & management, water & wastewater technology, operation & maintenance, and design, engineering & construction. Since North America is 19% of this market, the size of the project market is around \$36.70 billion [13].

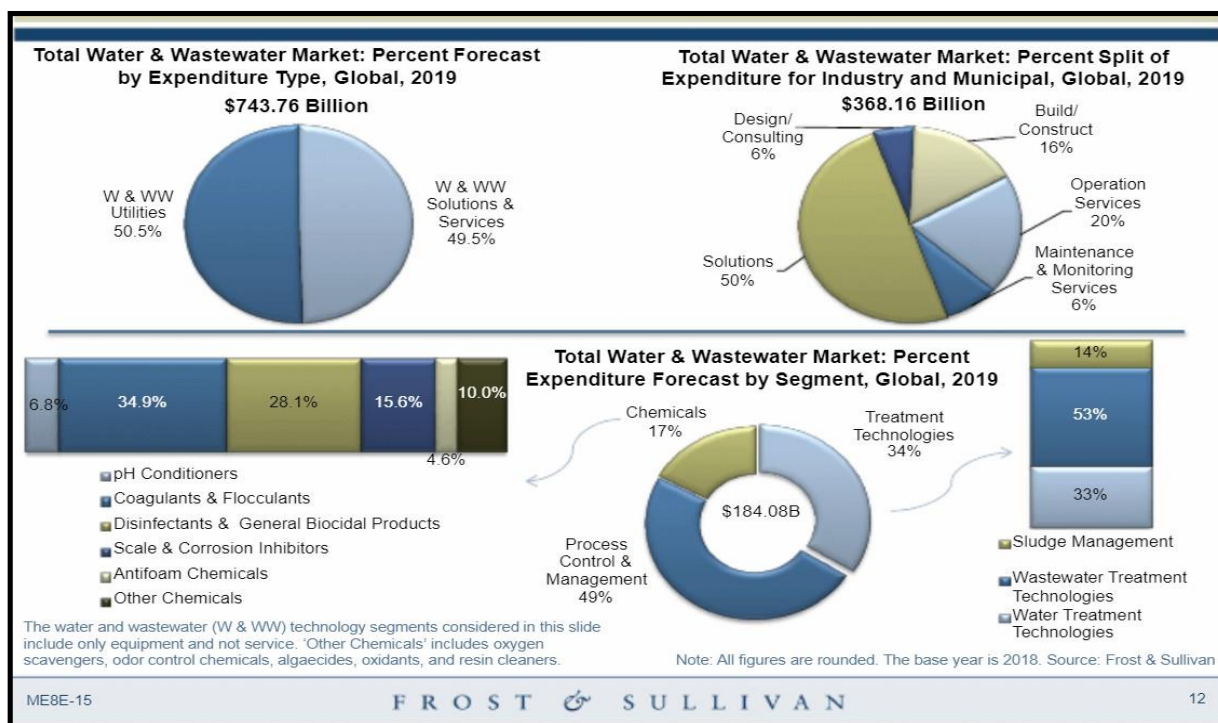


Figure 5: Total Water & Wastewater Market [13]

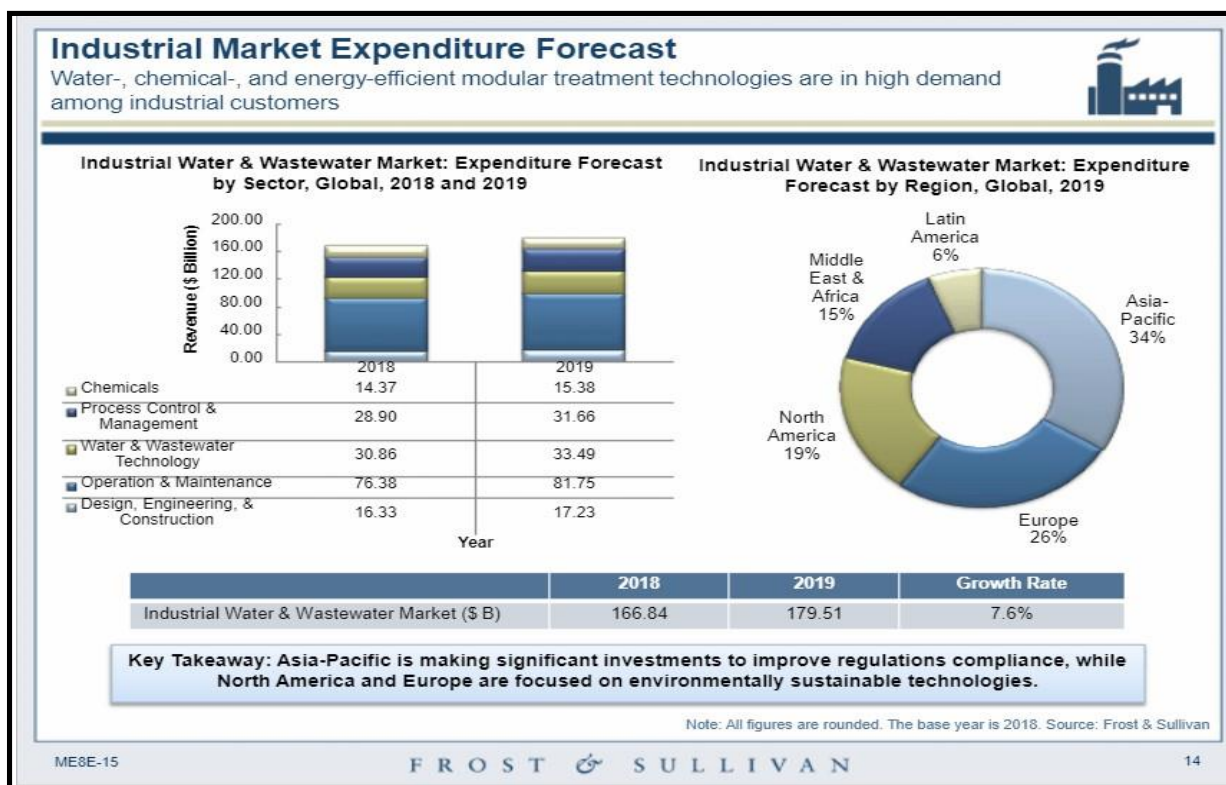


Figure 6: Industrial Market Expenditure Forecast [13]

The advantage of having a large market is that the technology continues to improve and helps to prevent further environmental effects. Such as leak detection using satellites and robotic sensors makes it economical to manage wastewater in more environmentally conscious ways. Henceforth, “rather than building treatment units at individual drilling sites or a large centralized treatment facility [one] can modularize the process. This has the advantage of reducing capital investment as well as the need to transport[wastewater] to a centralized treatment facility. This enables [the creation of a] flexible and adaptable framework for natural gas-producing companies to access on a site-by-site basis” [14]. The water treated can be assumed to have the same price of fresh water. It is treated to meet the same standard as that of the freshwater currently used to mix with the sand and the chemicals in order to create the fracking fluid that is the core of fracking.

Regulatory Environment

As time goes on the regulations for clean water get stricter however there are incentives to be more environmentally friendly. Departments such as Environmental Protection Agency (EPA), individual state’s Department of Environmental Protection (DEP), Occupational Safety and Health Administration (OSHA), and Department of Transportation (DOT) all are concerned with promoting safety and protecting the environment. Because water is a major essential to life on Earth some incentives have been established to offset the cost of removing contaminants from wastewater. For example, Pennsylvania’s DEP has the Pollution Prevention Program in which storage tanks are reimbursable up to the amount of \$2,500 per tank [15]. There are also energy incentives that may be applicable to implementing a mobile system in the form of grants, loans or rebates. By having a mobile treatment system rather than a fixed one, the CO₂ emissions from transporting water back and forth from the site in diesel trucks is reduced. This goes along with regulations such as the Clean Air Act of 1970 passed to regulate the air pollutants establishing the

minimum air quality standards [16]. By reducing vehicle traffic, the greenhouse gas emissions are reduced which help to stay on target. In the Marcellus Shale focus area of this project, “more than 350,000 oil and gas wells have been drilled in Pennsylvania since the first commercial oil well was developed in 1859” [17] and all aspects of water use and treatment have to meet standards mostly determined by the PA DEP.

Alternative Approaches

The main competitor to mobile wastewater treatment is a stationary water treatment facility using a conventional treatment process. The stationary treatment solution requires about 300 trucks per day to transport wastewater to the facility and that results in the release of about 40 tons of CO₂ into the environment. The trucks make multiple round trips a day which also affects the flow of traffic in the communities. Furthermore, “at the current price of diesel, significant cost impact is realized via this current philosophy. Off-site treatment is estimated to cost between \$0.03 and \$0.05 per gallon” [18]. Some advancements have already begun in making the treatment portable. “An estimated 50% - 60% total savings can be realized by treating and reusing flowback and produced water on-site” rather than having to transport to the flowback water to facilities elsewhere. As mentioned above in the value chain section, a major cost comes from transporting water to and from the site as well of obtaining the water. By having the onsite mobile treatment, the cost is reduced by about 75%. Therefore, the proposed approach is ideal for its environmental, cost, and regulatory benefits.

Process Description

Problem

The composition of the flowback water produced by hydraulic fracturing is complex, which makes it difficult to treat the wastewater. If this water leaks into the environment or is directly discharged without treatment, it will cause great ecological damage. Especially today, this is a critical concern because the world is facing a water shortage as explained above. The public is gradually becoming aware of the problems posed specifically by the wastewater produced by hydraulic fracturing. Storage and treatment of the wastewater both have their complications. Storage techniques such as open-air pits and deep-well injections lead to polluting the atmosphere and groundwater, but also lead to earthquakes. A series of improvements have been made to wastewater treatment methods. From the most primitive direct discharge steady progress is being made to approach the goal of zero liquid discharge technology for the wastewater produced by hydraulic fracturing. Common hydraulic fracturing wastewater treatment methods include the following.

Possible Solutions

Open Air Pits

Open air pits are a method of storing wastewater on surface impoundments. By using sunlight to evaporate the liquids in wastewater under sufficient light and heat conditions. Then the remaining sludge or solid waste was treated. This method can indeed reduce the cost of wastewater treatment, but it has high requirements on the local environment. Also, waiting for all the water to evaporate is a long process. Therefore, this method is not a universal method. This is only a rough collection of waste liquid; no treatment is performed to remove harmful substances. The disadvantages of this method are that it requires a sufficiently wide field to store the wastewater,

as well as being placed in an open environment. First, if natural disasters such as heavy rain or flood occur, the liquid can escape by overflow, by leakage. Another way is the containment dyke was damaged. Leakage can occur because the pit is unlined or because the lining gets damaged. Second, because it is an open environment, wild animals can gain access to drink the dirty water. These complex waters are likely to cause animal death. Third, because these wastewaters contain organic matter, they can easily volatilize into the air and cause local air quality to deteriorate, causing harm to the health of surrounding residents and workers.

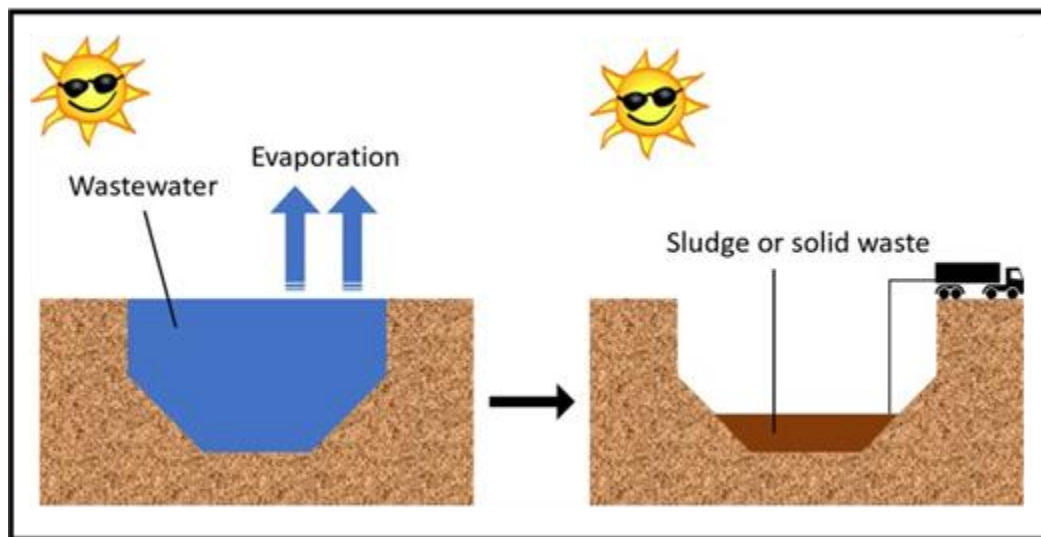


Figure 7: Scheme of Open-Air Pits

Deep-Well Injection

In the United States, underground deep-well injection of wastewater from hydraulic fracturing is one of the main disposal methods. As with Open Air Pits, the wastewater is pumped to the location untreated, the only difference is that the wastewater is injected underground. This method was first used by the oil industry in the 1930s, when produced brine from oil and gas wells was discharged back to shale formations. This simple and cost-effective method quickly became

widely used by industry. However, people soon realized that this could pollute groundwater and cause earthquakes. Earthquakes are developed as increased fluid pressure in a disposal migrates away from the well. Thus, this destabilizes pre-existing well-oriented faults [19]. As a result, the government began to develop regulations in hydraulic fracking disposals. The adoption of the Safe Drinking Water Act and the Federal Underground Injection Control (UIC) are the government's response in creating stricter laws for wastewater disposals.

According to the Underground Injection Control (UIC) program in the United States, underground injection wells are divided into 6 categories, of which the ones used to inject liquids related to oil and gas production are Type II injection wells. Class II wells have strict management procedures, their perfusion depth must be lower than the deepest underground drinking water aquifer. The construction, operation and monitoring of wells are also fully supervised. Nonetheless, some scholars worry that the movement of the formation may potentially contaminate drinking water. In order to protect underground drinking water, not all shale gas fields can use deep well injection to dispose of produced water. In some areas, underground disposal is prohibited because of geologic conditions. For example, in 2011, the Marcellus gas field produced 1,682,000 m³ of water, but there were only eight Class II wells in Pennsylvania that met the perfusion requirements. The underground perfusion capacity was significantly insufficient. Transportation to out-of-state installation greatly increases the cost of treatment of fracking flowback wastewater. [20]. In contrast, at the end of 2008, Texas had a total of 11,000 second-type perfusion wells approved by the U.S. Environmental Protection Agency. This total was slightly more than gas-producing wells and provided the disposal solution for the return fluid produced by shale gas development in the Barnett block [20]. Although, this is the cheapest method for treating hydraulic

fracturing wastewater, it has strict geological requirements for the disposal well and is subject to strict supervision by relevant authorities.

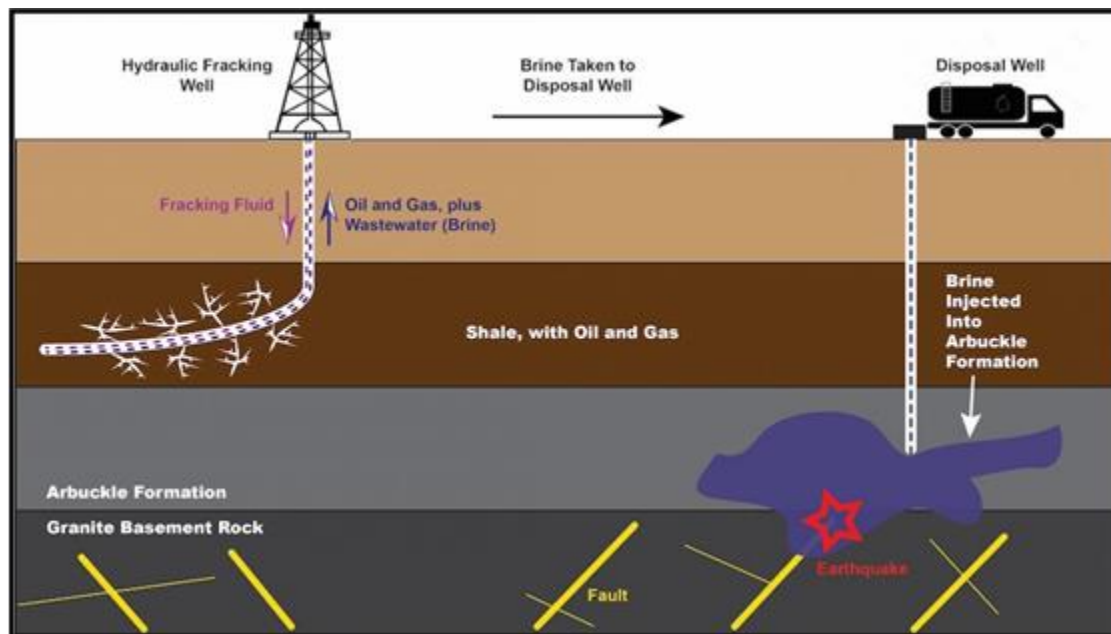


Figure 8: Scheme of Deep-Well Injection [21]

Off-Site Treatment

Off-site treatment is a method of transporting wastewater produced by hydraulic fracturing to other places for treatment. One method is to transport them to a municipal sewage treatment plant for treatment and discharge. According to statistics, the Marcellus shale area had more than 400 thousand m³ of hydraulic fracturing wastewater that was treated by the municipal sewage treatment plant [22]. However, because the municipal wastewater treatment plant process has almost no effect on the removal of total dissolved solids in the water, some surface water monitoring agencies in the Monongahela Basin have occasionally reported high salinity. Pennsylvania has therefore adopted stricter sewage discharge standards and management requirements. Since 2011, municipal sewage treatment plants that are located in the Marcellus

shale area are no longer accepting shale gas fracturing flowback water. Additional equipment is needed to make the municipal plants capable of treating flowback water, but the high capital cost has prevented the modifications from being made.

Another method of wastewater disposal is Bioremediation. Biological treatment methods are mainly divided into aerobic and anaerobic microorganism treatment. The commonly used activated sludge method and biofilm method are aerobic microorganism treatment methods - rotary disk aerobic reactor, SBR process, CASS process, and Bioreactor processing technology. Anaerobic microorganism treatment method requires the reaction to be performed under completely anaerobic conditions. For either technology, biological treatment of fracturing fluid requires pretreatment of wastewater and acclimatization and culture of bacteria in advance. The fracturing flowback water biodegradation process has a simple process flow, has the advantages of lower investment cost and simple operation, but it is not mature enough compared with other processes. Moreover, the biodegradation method takes a long time, and compared with the large amount of wastewater generated every day, it cannot meet the requirements of the processing speed.

In addition, one of the issues facing off-site treatment is transportation cost. One is transporting the wastewater by truck. The labor and fuel cost should be considered in this way. The average cost of it is about \$0.11/ton-mile. Another way is using pipeline. In this way, the company should pay the capital and operating cost. Its average cost is about \$0.5/bbl.

Chosen Solution

Our design solution to the problem of disposal of fracking flowback water is a mobile treatment system for fracking wastewater. The system will be modular on multiple trucks to be able to travel from fracking site to site throughout the Marcellus Shale area. A mobile treatment solution will make it possible to recover flowback water for reuse instead of disposal (because of hazardous chemicals that are returned from fracking). Our system will also eliminate the transportation cost to send water to an offsite treatment facility and is environmentally safer than wastewater storage. Our process includes 4 main stages; Oil removal, solid removal, organic removal, and salt removal. We plan to mount all stages on semi-trucks that can be moved from fracking site to fracking site to treat the flowback water from each fracking operation.

Hydraulic fracturing has a high probability of losing water that is initially injected. About 20-100% of the hydraulic fluid used can return to the surface, while the rest is stuck in shale deposits [25]. With that being said, any possibility of being able to reuse water, sand or additives will increase the likelihood of making a profit. As previously discussed, the solution of transporting the flowback water to a treatment site will only create additional charges. These charges can definitely lead to heavy costs if the fracturing site is far from any treatment facilities. Environmentally, a mobile system that treats wastewater at the fracking location is the best solution. Deep-well injection can lead to contaminants in groundwater and increase the likelihood of earthquakes. Storage in open-air pits also can lead to groundwater or soil contamination that may affect local farms, or air pollution from organic compounds in the wastewater evaporating into the atmosphere. Although leaks may occur in any process, it will be less likely for any leaks to occur at our on-site treatment process compared to the wastewater being transported using a vehicle.

Environment & Safety Concerns

Waste Solid & Liquid

The emissions in this process can be classified as brine, oil, and sludge. The brine contains high concentrations of sodium, calcium, barium, and chloride ions. Because it has the characteristics of high salt content and high toxicity, if it is directly discharged, it will cause significant harm to the natural environment. Improper handling of it may cause the pollution of underground water and the increase of salinity. Concentrated brine will settle to the bottom after draining into the water, which will consume dissolved oxygen in the receiving water body. High salinity and reduced dissolved oxygen levels can have profound effects on benthic organisms, which in turn produce conduction effects throughout the food chain. If it is discharged on the land, it will cause soil salinization, compaction, and other problems, which will damage local plant growth. Because the salinity of high-salt wastewater cannot be removed by natural biodegradation processes, it must be properly disposed of.

The main components of oil are alkanes and aromatics. If the oil floats on the water, it will quickly spread and form an oil film. Oil can stick to fish gills. It can also suffocate fish, inhibit waterfowl from spawning and hatching, destroy the impermeability of their feathers, and reduce the quality of aquatic products. Oil film formation can hinder the reoxygenation of water bodies, affect the growth of plankton, and disrupt ecological balance. When it enters the soil, it is difficult to grow plants in the contaminated land for a long time because it is difficult to degrade. At the same time, if the groundwater is contaminated with harmful substances, it will enter the food chain and eventually cause harm to humans.

The sludge composition is complex. In addition to containing a large amount of water, it also contains difficult-to-degrade organic matter, heavy metals, salts, and a small number of

pathogenic microorganisms. If not handled properly, sludge entering the environment will not only cause serious secondary pollution. The salt in the sludge will directly affect the electrical conductivity of the soil, disrupt the plant nutrient balance, inhibit the plant's absorption of nutrients, and even cause direct damage to the plant's root system, leading to plant death. Moreover, the antagonism between the salt isolates will accelerate the leaching of effective nutrients in the soil, disrupt the soil nutrient balance, and make the soil compact and barren. A large number of bacteria, viruses, and parasite eggs contained in sludge cannot be ignored. They must be prevented from entering the food chain to reduce their harm to human and animal health. At the same time, some heavy metal ions and radioactive substances contained in it will also enter the sludge, which must be prevented from entering the environment. Radioactive pollution is difficult to eliminate, and the intensity of radiation can only weaken over time. Under special circumstances, radioactive elements may be contaminated by food through animal or plant enrichment, causing harm to human health [33].

Air Emissions

The gas emissions here are mainly low-boiling organic compounds. One type is that the sun will heat lagoon, and its pressure is lower than the API separator, so some low boiling hydrocarbons will evaporate from the liquid, including benzene, toluene, etc. When these organic vapors accumulate to a certain degree, it will cause certain health damage to the people around them, and even cause cancer. At the same time, this will also have a certain impact on the surrounding vegetation. When these gases enter the atmosphere, they cause a large amount of ozone to decompose, triggering the greenhouse effect, acid rain, and ozone layer damage.

Mitigate the Effects of Discharges

Reducing emissions is the best way to achieve pollution prevention. The development of complementary environmental protection strategies based on waste reduction represents a significant change in thinking. Reducing waste is an obvious way to offset the increasing economic and environmental costs of managing large amounts of waste. It also addresses concerns about economic inefficiencies caused by increasingly stringent pollution control regulations. Waste reduction refers to in-factory practices that reduce, avoid, or eliminate the generation of hazardous waste, thereby reducing health and environmental risks. The goal of waste reduction is to change practices and design future processes and operations to reduce the degree of waste hazards and the amount to be managed, controlled, and regulated. Preferred methods of dealing with hazardous waste include [33]:

- waste reduction
- waste prevention
- waste minimization/avoidance
- waste abatement
- waste elimination

Hazardous waste refers to all non-product hazardous outputs from industrial production to all environmental media, even though they may be within permitted or permitted limits. To effectively prevent chemical hazards, two aspects should be considered. First, reduce emissions. Second, strengthen governance. Governance includes proper disposal, recycling, and resource utilization of waste.

Improving production equipment is an important way to achieve clean production and control of pollution sources. If direct cooling of chemical substances is changed to indirect cooling, the emissions of pollutants can be reduced. In addition, improving the tightness of equipment and

pipelines, strengthening the management of enterprises, improving the quality of operators, reducing the leakage of raw materials and products, will also reduce the degree of pollution. Adopting a closed circulation system is a new development direction for pollution prevention in chemical production. The so-called closed-cycle means that the waste in the production process is returned to the production system for use through a certain treatment technology to prevent pollutants from being discharged into the surrounding environment while improving the utilization of raw materials and increasing the product yield. This system can reduce the raw material consumption quota and reduce the harm of pollutants.

In the chemical production process, it is vital to ensure safe production. Accidents in chemical production will cause not only serious human and economic losses to the factory itself, but also cause great pollution and harm to the surrounding environment. Therefore, strengthening safety education and management in the process of chemical production to prevent accidents is also an important way to prevent chemical pollution [33]. The design for each specific unit is as follows.

Table 1: Equipment & Features

Equipment/Process	Potential Hazard	Design Features
V-101	Gases Vent during filling, possible spills, Elevated platform - fall risk, confined space, Overpressure risk (from letdown valve failure)	Pipe to safe location, safety rails, safety procedures, pressure relief valve
Lagoon	Evaporation to atmosphere, possible spill, personnel exposure	Monitor environmental quality regularly, fencing/barrier and signage around lagoon
Pumps	Seal can leak, Rotating parts, electricity	Periodic check, mechanical guards, explosion proof design, electrical protection
Flocculation		
T-101	Carbon dust is explosion hazard, can absorb oxygen during loading, elevated platform, fall risk, confined space, Gases vent during filling	Detect possible spills, TCLP (Toxic chemical leaching potential test) test required, send to treatment plant
V-102 Centrifuge	Rotating parts, electricity, seal can leak	Signage, mechanical guards, insulation, seal design features, electrical trips and safeguards
Diesel generator	Hot surfaces, noise, rotating parts, electricity	Signage, muffler, noise shielding, remote location, PPE, electrical trips and safeguards.
Truck	Noise, seal can leak, static electricity	Remove static electricity, signage
Filling trucks	Gases vent during filling, possible spills	Periodic check, secondary containment and spill kits are required
F-103 - F-106	Fouling or wear	Exposure and emissions during periodic membrane cleaning/replacement

There are no OSHA PSM highly hazardous materials in this process

Toxicity

Based on European Union (EU) guidelines, the toxicity can be classified in Table 8.

Table 2: Classification of Toxicity

LD50 Absorbed Orally in Rats, mg/kg	Classification
≤25	Very Toxic
25 to 200	Toxic
200 to 2000	Harmful

The data for LD50 can be found by MSDS for the following materials:

Table 3: Toxicity of Materials [34,35]

Compound	LD50 (mg/kg)	Classification	PEL	Carcinogenicity
Benzene ¹	930	Harmful	1 ppm TWA, 5 ppm STEL	Group 1: Carcinogenic to humans
Ethylbenzene	3,500	-	100 ppm TWA, 125 ppm STEL	-
Styrene	> 5,000	-	50 ppm TWA, 100 ppm STEL	Group 2A: Probably carcinogenic to humans
Toluene	2,600-7,500	-	10 ppm TWA, 150 ppm STEL	-
Xylene	3608	-	100 ppm TWA, 150 ppm STEL	-
Pb (VI)	1947	Harmful		“Reasonably anticipated to be human carcinogens”
Ba (II)	150	Toxic		-
Cr (VI)	1870	-		Group 1: Carcinogenic to humans
Nitrite	150	Toxic		Group 2A: Probably carcinogenic to humans

Operators or people in close contact with hazardous substances should wear personal protective equipment. Standard PPE in process plants is safety glasses, steel toed shoes, gloves,

fire-resistant clothes, hearing protection, and a hard hat. Auxiliary PPE is respirator, dust mask, face shield, specialized gloves, apron, and more when needed. Personal protective equipment may include items such as gloves, safety glasses and shoes, earplugs or muffs, hard hats, respirators, or coveralls, vests and full body suits [36].

Explosive

Table 4: Flammability Ranges [34]

Compound	Lower Limit	Upper Limit	Flammability (NFPA 704)	Boiling point (°C)	Flash point (°C)	Autoignition temperature (°C)
Benzene	1.3	7.9	3	80.1	-11.63	497.78
Ethylbenzene	1	7.1	3	136	22.22	430
Styrene	1.1	6.1	3	145	31	490
Toluene	1.1	7.1	3	111	6	480
Xylene	0.9	6.7	3	144	32	463
Ethane	3	12.4	4	-88.6	-135	472
Methane	4.4	16.4	4	-161.5	-188	537
Propane	2.1	10.1	4	-42.25 to -42.04	-104	470
Diesel	0.6	7.5	-	180 to 360	52 to 96	210

Because some compounds are flammable and explosive and equipment is next to a gas well, it requires the plant to have appropriate explosion-proof facilities throughout our process. The metal enclosure of electrical equipment should be grounded. The electrical and instrumentation equipment in the installation area is pressure-proof and explosion-proof. Installation and maintenance equipment in the installation area requires the use of special explosion-proof tools made of special materials, such as beryllium copper alloy explosion-proof

tools. It does not cause friction and impact sparks and does not even generate hot high-temperature surfaces.

In addition, in the area of process devices and storage and transportation facilities that generate flammable and toxic gases during the production of the device, it is necessary to have benzene monitors and flammable gas detectors to monitor possible leaks.

(1) Combustible gas detection and alarm devices shall be set, and primary alarm and secondary alarm shall be adopted. At the same time as the secondary alarm, the contact signal is output for the interlock protection system; meanwhile, continuous detection, indication and alarm are performed. Record or print the alarm.

(2) The control room, power distribution room and on-line analysis instrument room in trailers should be received with the signal from the combustible gas concentration detectors in the field near potential sources of leaks.

(3) The instrument should choose anti-static and explosion-proof type which meet the “Class 1 Division 1” design requirements.

(4) All instrument alarm signals should be sent to the control room uniformly.

(5) Periodic calibration and maintenance of each meter

(6) Equipment/vessel overpressure protection by relief valves piped to flare. Electricity-powered equipment shall Meet Class 1 Division 2 requirements of the National Electrical Code. Benzene monitors required to prove personnel exposure <0.5ppm. Operators shall have eye protection, steel-toed boots, hard hats, and Nomex fire-resistant clothing.

HAZOP Analysis

A HAZOP safety analysis for the submerged microfilters, settling tank, decanter, and backwash tank are displayed in Tables 5 to 8. The purpose of the equipment mentioned is to ensure an efficient process in removing solid material from flowback water. All this equipment plays a critical role in solids removal. The settling tank and decanter control the input process streams and control the flowrate and compositions prior to entering the filters. The backwash tank is used in storing clean effluent water to be used in each backwash cycle to clean the filters and prepare them for the next cycle. The microfilters inlet pressure is 130 psi, with an expected pressure drop of around 30 psi. The solids must be removed to prevent plugging of the carbon absorber bed or fouling of the RO membrane equipment downstream. The backwash cycle will ensure a proper cleanse in the filter's membrane that will result in a higher yield of removing solids.

Table 5: HAZOP Analysis for Settling Tank (TK-103)

Settling Tank: TK-103					
Intention: The sedimentation tank allows suspended particles to settle out of water					
Guideword	Parameter	Cause	Consequence	Safeguard	Action
Line "Effluent Stream" From TK-102					
Intention: The slow mixing tank flow is passed on to the sedimentation tank					
NO	FLOW	Pipe Blockage/Leakage/P-101 Failure/Lose Well Flow	Halt to operation can cause delay	Lagoon has the surge capacity for interrupted operations such as maintenance and regeneration	Existing safeguards are adequate
LOW	FLOW	Problem with P-101/ Level control indication	More time for settling and cleaner product	Level instrument controls pump speed /Instrument calibration	Regular pump maintenance and instrument calibration
HIGH	FLOW	Problem with P-101/ Level control indication	Lack of proper retention time/premature filter regeneration/ System capacity decreases	Level instrument controls pump speed /Instrument calibration	Regular maintenance on valves, pipes, and pump
Line "Overflow Stream"					
Intention: The overflow is sent to P-102 to then be sent to filters to continue purification					
Guideword	Parameter	Cause	Consequence	Safeguard	Action
LEAK	LOSS	Corrosion/ P-102 seal failure	Exposure to flowback water and its contaminant	Material Selection and good engineering design	Regular maintenance and inspection
NO	FLOW	Filter Blockage because delta P instruments are indicating incorrectly	Possible leak and loss of production	Automated filter regeneration based on differential controller	Regular maintenance and inspection
LOW	FLOW	Filter Blockage because delta P instruments are indicating incorrectly	Possible leak and loss of production	Automated filter regeneration based on differential controller	Regular maintenance and inspection
HIGH	FLOW	Problem with P-101/ Level control indication	Faster solid build up in filters which decreases the regeneration time	Variable speed drive on P-102 based on the flow	Regular maintenance and inspection
Line No. "Underflow Stream"					
Intention: The tank's particle that settled at the bottom with some water are sent to the decanter to removed more solids, passes through P-103					
Guideword	Parameter	Cause	Consequence	Safeguard	Action
NO	FLOW	Pipe Blockage/Leakage/P-101 Failure/Lose Well Flow	Pump may run dry	Variable speed drive on P-103 based on the level of TK-103	Regular maintenance and inspection
LOW	FLOW	Problem with P-101/ Level control indication	Less product needed to be filtered which help the filtration system	Level instrument controls pump speed /Instrument calibration	Regular pump maintenance and instrument calibration
HIGH	FLOW	Problem with P-101/ Level control indication	Can surpass the decanter's capacity and cause a shutdown	Level instrument controls pump speed /Instrument calibration	Regular pump maintenance and instrument calibration

Table 6: HAZOP Analysis for Decanter Centrifuge (V-102)

Decanter Centrifuge: V-102					
Intention: Removing solids from water by rotating a bowl at high speeds					
Guideword	Parameter	Cause	Consequence	Safeguard	Action
HIGH	FLOW	Flow from underflow stream: increase suction pressure by the pump	Overflow in the decanter	Variable speed control on P-103 to from level control on TK-103	Install alarm
	FLOW	Decanter did not separate the sludge from water properly	Decanter structure blockage	Motor control on the unit, VSD AIC AE	Regular maintenance and inspection
	SPEED	Faulty motor/ High solids content	High amount of water lost with the sludge	Motor control on the unit, VSD AIC AE	Regular maintenance and inspection
LOW	SPEED	Faulty motor/Wrong detection of solids	Decanter Shutdown	Motor control on the unit, VSD AIC AE	Regular maintenance and inspection
	FLOW	To the sludge truck: Lack of settling time on TK-103/Wrong transmitter reading	Less production of the dewatered sludge	Level instrument controls pump speed /Instrument calibration	Regular pump maintenance and instrument calibration
	FLOW	To the filter: Lack of settling time on TK-103/ Wrong transmitter reading	Less product needed to be filtered which help the filtration system	Level instrument controls pump speed /Instrument calibration	Regular pump maintenance and instrument calibration

Table 7: HAZOP Analysis for Backwash Tank (TK-104)

Backwash Drum: TK-104					
Intention: Holds storage of effluent for backwash system					
Guideword	Parameter	Cause	Consequence	Safeguard	Action
HIGH	Pressure	Pressure Control Failure	Can exceed vessel pressure limit - eruption	Pressure relief valve at discharge pipe of lagoon	Adjust discharge pressure of pump P-104
LOW	Level	Tank Leakage	Pump failure once backwash system initiated	Material Selection and good engineering design	Backwash system stop - shutdown of backwash pump P-104

Table 8: HAZOP Analysis for Microfiltration (F-101/2)

Microfilters: F-101/102					
Intention: Removal of inorganic solids					
Guideword	Parameter	Cause	Consequence	Safeguard	Action
NO	Pressure Drop	Membrane Failure/ corrosion of the membrane support	Solid leakage into downstream equipment/ water leakage into sludge	Monitor filter differential pressure	Membrane replacement/DPAL - Differential pressure alarm low
HIGH		Accumulation of Particles in Membrane	Filter element replacement/ higher filter pressure drop which increases the regeneration frequency/lowers the flowrate	Monitor filter differential pressure	Membrane replacement/DPAH - Differential pressure alarm high
		Aging of the Membrane (solids stuck permanently)	Filter element replacement/ higher filter pressure drop which increases the regeneration frequency/lowers the flowrate	Monitor filter differential pressure	Membrane replacement/DPAH - Differential pressure alarm high
		Solid characteristic change (i.e. finer particles, compressibility, cohesiveness)	Filter element replacement/ higher filter pressure drop which increases the regeneration frequency/lowers the flowrate	Monitor filter differential pressure	Membrane replacement/DPAH - Differential pressure alarm high
LOW	Flow Rate	Flow instrument calibration off	Pump Speed too low	Instrument calibration and maintenance/ use a low fouling instrument (i.e. Coriolis)	Recalibrate and clean the instrument
		Regeneration failure	Loss of production	Automated system bases on delta pressure	Filter maintenance
		Fouling	Filter element replacement/ higher filter pressure drop which increases the regeneration	Monitor filter differential pressure	Offline Cleaning
HIGH		Flow instrument calibration off	Pump Speed too high	VSD on Pump 102/103	Lower speed on Pumps
LEAK	LOSS	Valve seal leak/corrosion of equipment	Exposure to flowback water and its contaminant	Material Selection and good engineering design	Regular maintenance and inspection

Process Control

Activated Carbon Adsorption Column (T-101)

The adsorption column is used to remove dissolved organic material from the flowback water using activated carbon. The organic molecules from the water solution will be drawn into the porous surface of the activated carbon. To ensure an efficient column, the amount of contact time the carbon has with the solution is highly important. Therefore, two pressure transmitters will be placed before and after the column to monitor the pressure drop. From here, the transmitters will read the pressures and sent the signal to the delta pressure controller. Once delta pressure controller reads a pressure that's greater than 35 psi, a high delta pressure alarm will go off. A high-pressure drop will indicate that the activated carbon is not plugged. Whereas a low-pressure drop will express a possible disturbance in the column. Exhausted carbon within the column will no longer purify the solution. Thus, the alarm will be there to notify an operator that the carbon has been exhausted and to replace the column with a fresh supply.

Pump 105 also plays an important role in pumping at a high enough pressure of 50 psi. When it comes to activated carbon adsorption, the higher pressure within the column will result in a greater adsorption capacity. Pump 105 can also be used in regulating the flow rate into the column. Regulating the flow rate is essential in order to not exceed the flux of the activated carbon. The control objective would be to keep the inlet flow constant in order to achieve the maximum capacity. This will lead to saving any cost of replacing the carbon. The volume of carbon needed in the column will be constant, as that was hand calculated based on the contact time and expected volumetric flow rate. Exceeding the mass flux can result in inefficient adsorption that can lead to organics breakthrough, damaging the RO membranes, or excessive consumption of carbon.

Table 9: T-101 Control Variables

Control Variable	Set Point	Lower Limit	Upper Limit	Manipulated Variable
Delta Pressure	10 psi	5 psi	15 psi	Variable Speed Drive (P-105)
Flow	21,000 gal/hr	15,000 gal/hr	27,000 gal/hr	Variable Speed Drive (P-105)

Settling Tank (TK-103)

The settling tank uses gravity in separating particles from liquids after electrocoagulation and flocculation. Level is the main variable we need to control on the tank. Different than the other tanks in process, the settling tank will need two level transmitters, one for the sludge and one for the liquid level. To measure the sludge level, the hydrostatic pressure will be measured from the top and bottom of the tank. The level transmitter will then send a signal to the level controller. The level controller will then manipulate the variable speed drive of pump 103 to transfer sludge to the decanter. The liquid level transmitter will use a float type level instrument for the upper level measurement. The level transmitter will send a signal to another level controller, that will then manipulate the variable speed drive of pump 101 located at the lagoon. Pump 101 will speed up or slow down to deliver enough water to the process to keep TK-103 liquid full, while pump 103 will control the rate of sludge exiting the tank to maintain a constant solids concentration in the decanter feed. Only P-101 has the functionality in preventing overflow in the settling tank. Since the settling tank is planned to be inside a truck next to the flocculation and electrocoagulation tanks, an overflowing tank can lead to an environmental problem that will have to be cleaned up and reported to the Environmental Protective Agency depending on the contents of the water. The last process element that contributes to the settler's tank's preformance is pump 102. This pump is used in regulating the liquid flow to the microfilters, which require a constant flow for best performance. A flow transmitter is placed at the discharge of the pump, then that transmitter sends a signal to flow controller that manipulates the variable speed drive of the pump to regulate the flow of the discharge stream.

Table 10: TK-103 Control Variables

Control Variable	Set Point	Lower Limit	Upper Limit	Manipulated Variable
Solid Level	3 ft	4 ft	6 ft	Variable Speed Drive (P-103)
Liquid Level	9 ft	8 ft	10 ft	Variable Speed Drive (P-102)

Microfiltration and Backwash System (F-101/2 & TK-104)

The submerged microfilters are used in filtering out particulate materials from the flowback water. One of the essential process control elements is pressure drop across the filters. As mentioned in the HAZOP, disturbances such as ageing, failure, or accumulation of particles in the membrane can occur. Therefore, a high pressure drop across the filters can give us an alert of these possible issues. A delta pressure cell will be used in order to directly measure the pressure drop accurately across each filter. These delta pressure instruments will send a signal to the delta pressure controller. Each filter will have its independent delta pressure controller in order to read two different pressure drops across each device. The delta pressure controller sends a signal to the backwash controller to initiate a backwash cycle when the filter ΔP reaches about 35 psi, indicating that enough solids have accumulated to require a regeneration. A high delta pressure alarm will be placed at each controller in case a high-pressure drop is still occurring after a completed backwash. This will enable early identification of problems such as aging of membrane that will need to be replaced offline.

Backwash for the microfilters removes any accumulating particles in the membrane using a reverse flow. Once a signal is delivered from the delta pressure controllers that there has been a pressure drop greater than 35 psi, the backwash system will begin. The backwash controller will initiate a shutoff of the process pumps feed and will close valves V-101 to V-102. Next, the backwash controller will start the backwash pump (P-104) on spillback and manipulate the pump's

variable speed drive. Then, the backwash controller will open valves BV-101, BV-102, and BV-103. The backwash system will proceed at a constant flow for about 10 minutes to ensure most particles are removed from the membranes medium. Once the backwash streams exit the filter, it'll lead back to the lagoon where that backwash water will be cleaned in the same normal procedure. After the fixed 10-minute time is over, the backwash controller will shut off the backwash pump, open the backwash inlet and outlet valves, open process outlet and inlet valves for process, and restart flow through the filter. The backwash controller will run from a range of 15 to 23 gallons per minute per square foot [37].

Other controllers that are a part of the process loop include valve BV-105. The flow rate of the discharge of the pump is sent to the flowrate transmitter, which sends a signal to the flow controller. This then manipulates that valves' s opening position to maintain sufficient flow to dislodge the accumulated solids from the filter media. Valve V-105 prevents overpressure of the filters during backwash by spilling back pump 104 discharge to the backwash tank on pressure control. A pressure transmitter is placed at the discharge of the pump 104, then sends a signal to the pressure controller. The pressure controller than manipulates V-105's valve positioning in order to regulate the amount of pressure going to the filter. Lastly, decanter centrifuge measures the density of it's effluent. The density transmitter sends the signal to the density controller, which then manipulates the variable speed drive of the motor for the bowl and the scroll for the decanter. The decanter centrifuge is used to recover more of the water from the sludge from the settler and create a more efficient process for F-101.

Table 11: Control Variable for F-101/2 & TK 104

Control Variable	Set Point	Lower Limit	Upper Limit	Manipulated Variable
Delta Pressure (Filters)	25 psi	10 psi	40 psi	Valves & Variable Speed Drive (P-104)
Density	8.42 lb/gal	8.3 lb/gal	8.53 lb/gal	Variable Speed Drive (Motor)
Pressure (TK-104)	90 psi	85 psi	95 psi	Valve Position
Backwash Flow	19 gpm/square ft	15 gpm/square ft	23gpm/square ft	Valve Position

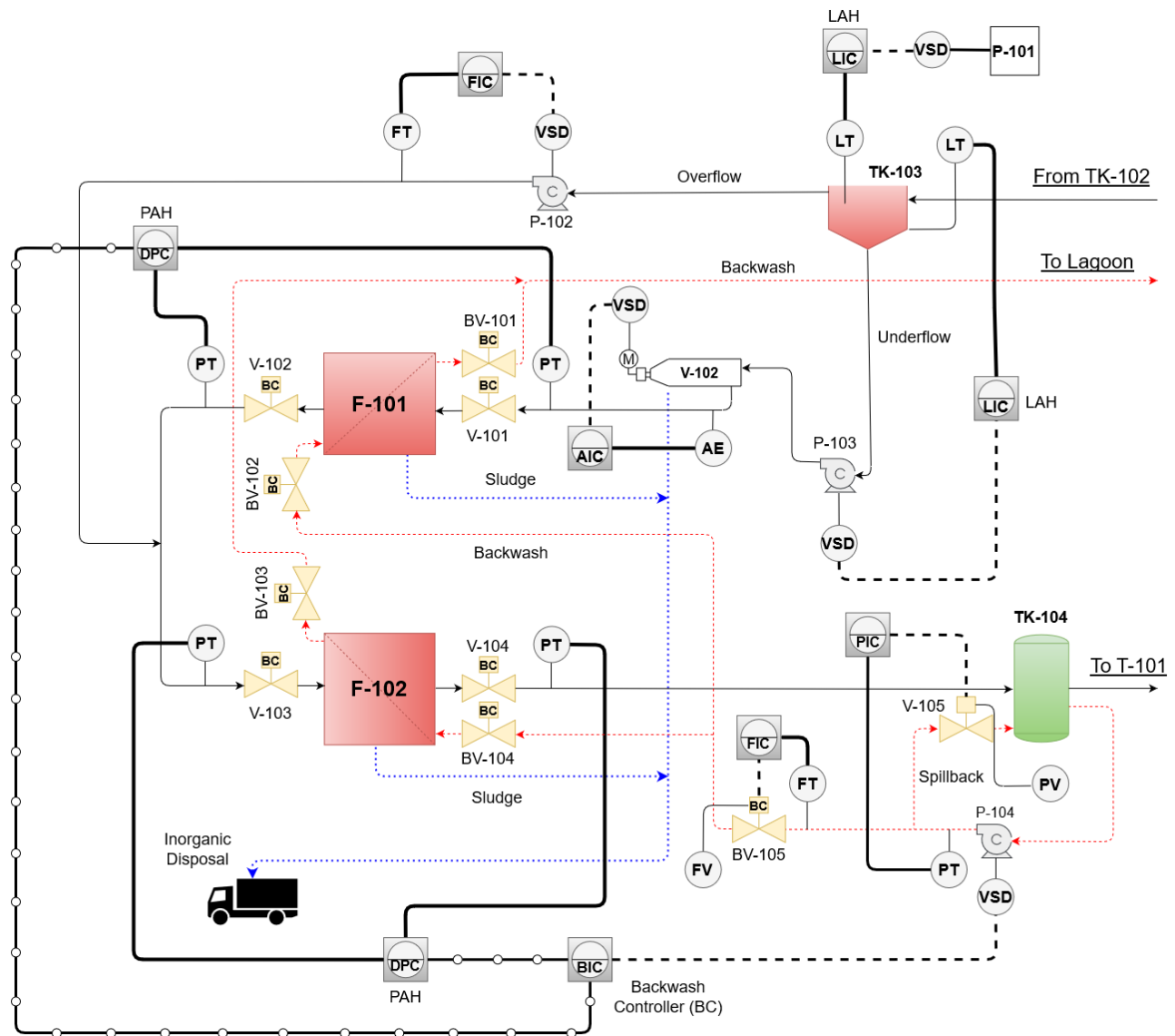


Figure 9:Control Loop for Microfiltration

Economic Analysis

The wastewater treatment will operate 330 days of the year. Each site will be given an estimate of six days to treat their flowback water. A day and a half will be for travel time, another day and a half for set up and take down, and the other three days will be for treating the water. There will be a total of three shifts and each shift will have 4 workers being paid \$20.05 per hour [38]. In a year, the operating labor would be \$1.92 million. The reason we only have 12 workers is because the equipment will be inside commercial trailers and the cost for transporting the trailers to the designated fracking sites will be from hiring a third-party trailer driver. The trailer transportation is further discussed in the VCOP section. The supervision cost would be 25% of the operating labor [39] to total a \$481,200 cost. The direct salary overhead and general plant which are 45% of the operating labor and supervision and 40% respectively, will sum up to be \$2.05 million. Maintenance, insurance, environmental charges, and capital charges have their corresponding percentages from certain operating costs as shown in the table in Appendix and they sum up to be \$383 thousand. The reason royalties and rent of land are zero because this is not permanent installation, the mobility ability is responsible for eliminating those charges. The equipment can be stored in a rented facility whenever it is not on a site as well as when it need maintenance. The last cost would be for sales and marketing to be able to promote the treatment and continue to be able to operate, the cost is 5% of the total cost of production which would accumulate to be \$411 t. The figure below shows the distribution of the mentioned costs which are the Fixed Cost of Production (FCOP).

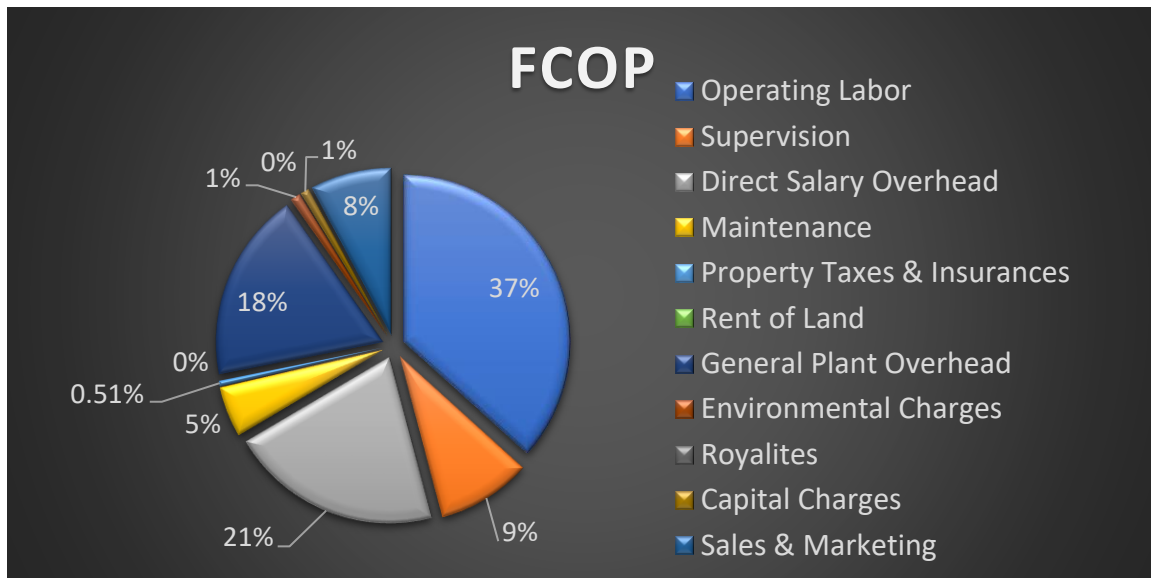


Figure 10:Fixed Cost of Production

The Inside Battery Limits (ISBL) of the mobile treatment, which includes equipment cost that were obtained from ASPEN simulation, ten trailers needed to fit the equipment [40] , the installation of the equipment in the trailers, and the diesel power generator [41] used to operate the treatment has a cost of \$5.13 million. The treatment consists of four different water purification processes which are, oil removal, solid removal, salt removal, and organic removal. The organic removal accounts for 72% of the ISBL costs. The percentages of the total ISBL for each process is in Figure 10 below. The organic removal accounting for almost 75% of the cost stands out and signals a consideration to determine any way the cost can be reduced. The activated carbon consumption could be lowered, find a better supplier with cheaper prices, or possible increase the carbon capacity.

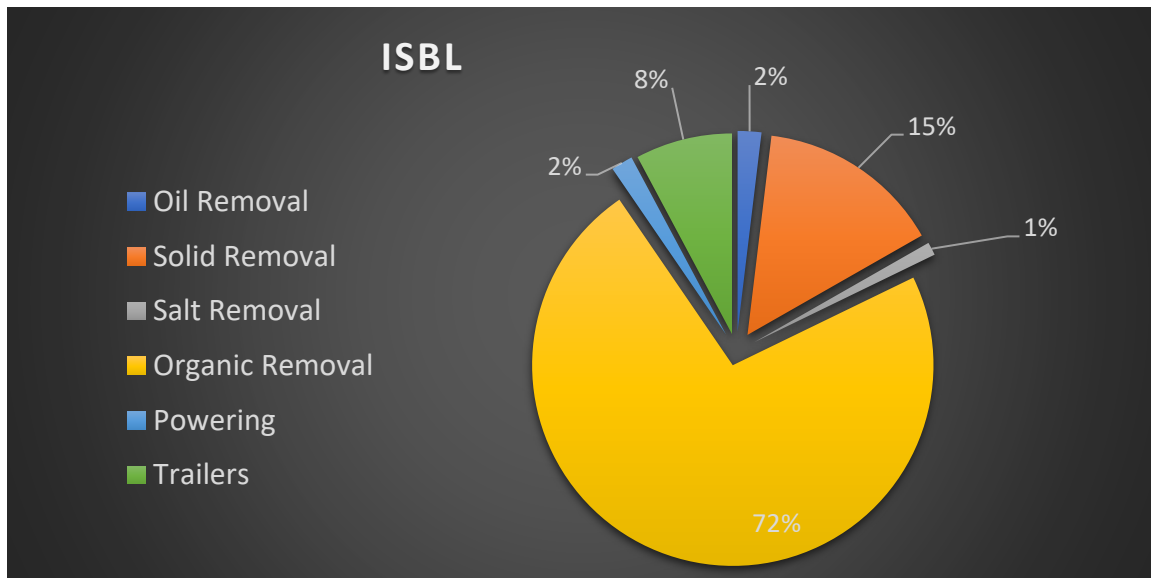


Figure 11: Inside Battery Limits Breakdown

The total for the Offsite Battery Limits (OSBL) of this process is based off the ISBL calculated. The OSBL is 5% of the OSBL [39] and was found to be a total of \$256 thousand . A contingency fund for unforeseen circumstances was a total of \$538 thousand. Engineering and construction are a total of OSBL \$1.03 million based on them being 20% of the ISBL and OSBL sum. The working capital is 15% of the ISBL and that amount to a total of \$769 thousand. Summing up all the discussed costs the Total Capital Cost is \$7.71 million.

The treating of the flowback water is prices at 15 cents per gallon [42]. Each site has an average of 1.5 million gallons of flowback water and in the Marcellus Shale there is an average of 120 new wells drilled each month [43]. As mentioned before each site is allocated to have six days, by operating the 330 days, 55 sites could be visited to provide to the service. This would amount to \$12.4 million per year. Another source of revenue is the water itself; each site has an average of 1.44 million gallons that would be produced as clean water. This water would be sold to them at a 30% cost of the freshwater cost to ensure both parties benefit. The cost of water is \$0.0083 per

gallon [42]. The total revenue per site from this would be \$3,600 which that multiplied by the 55 sites equals a revenue of \$199,000. The third source of revenue is from organic solids being collected and sold such as benzene, ethylbenzene, styrene, toluene, ethane, methane, and propane. These generate \$4,600 dollars in revenues since a gallon costs \$6.2 [44] and each site makes about 909 gallons. The sum of the revenues is \$12.61 million each year. Figure 11 below displays the breakdown of each of the revenue's contribution to the total.

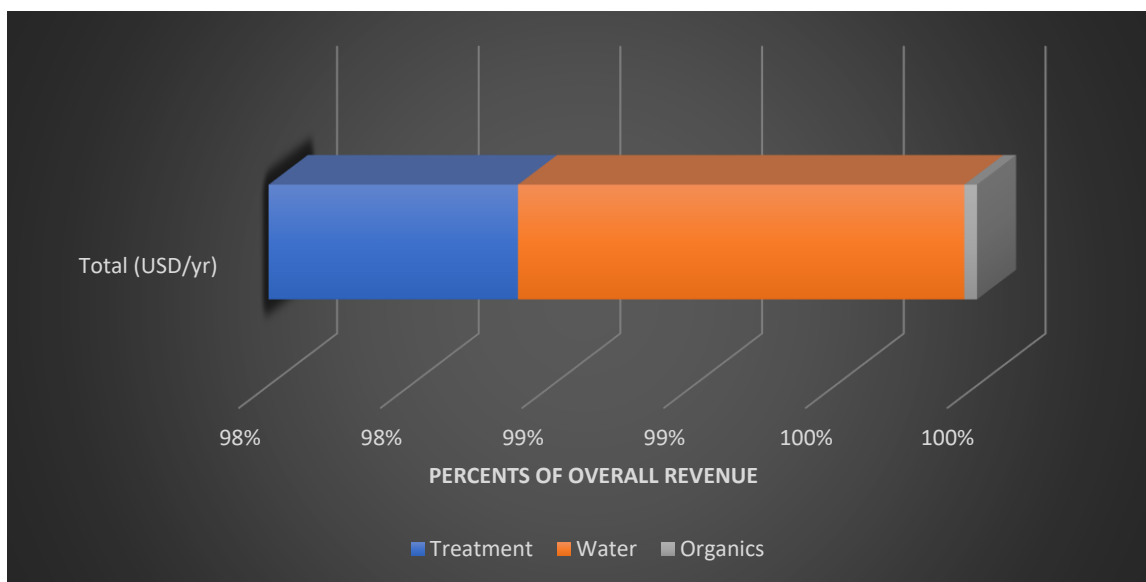


Figure 12:Revenues

The Variable Cost of Production (VCOP) is associate with the size of the company and treatment of the water. The only raw material present is the activated carbon used in the adsorption column. This requires 8.2 tons per day and each ton costs \$2,000 [45], the total amount of carbon needed is \$2.71 million. The only utilities come from the diesel fuel for the engine [46], the cost is \$2.85 per gallon and the use is 16.10 gallons per hour to equal \$181,700. For the disposal, the brine will be deep well injected and the cost for that is \$0.054 per gallon [47], in a year \$197,000 will be spent on that disposal. The sludge will be disposed of with the landfill [48] and the cost is \$8.35 per ton to equal a total of \$135,000 each year. The transportation cost is from hiring a third

party to pick up and drop off the trailers. The cost is \$3.39 per mile, and on this is from the average that the distance is 100 miles per truck per site which equals \$159,330 each year. Figure 12 below shows the breakdown of the VCOP which it's total cost per year is \$3.38 million.

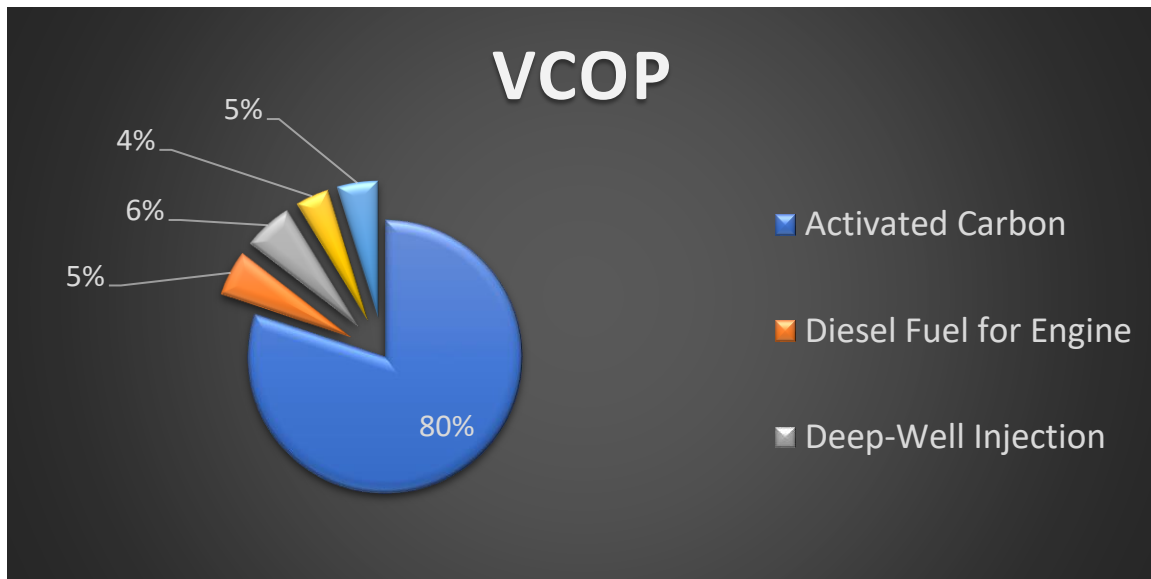


Figure 13: Variable Cost of Production Pie Chart

The Total Investment for this project is the sum of ISBL & OSBL, Contingency which is 10% of the sum of ISBL and OSBL. Plus, the sum of Engineering and Construction which are 10% of the sum of ISBL and OSBL. And the last one is the working capital which is 15% of the ISBL. These percentages were taken from the Gavin Towler book. Figure 13 below displays the percentage distribution for each of the items what make up the total investment.

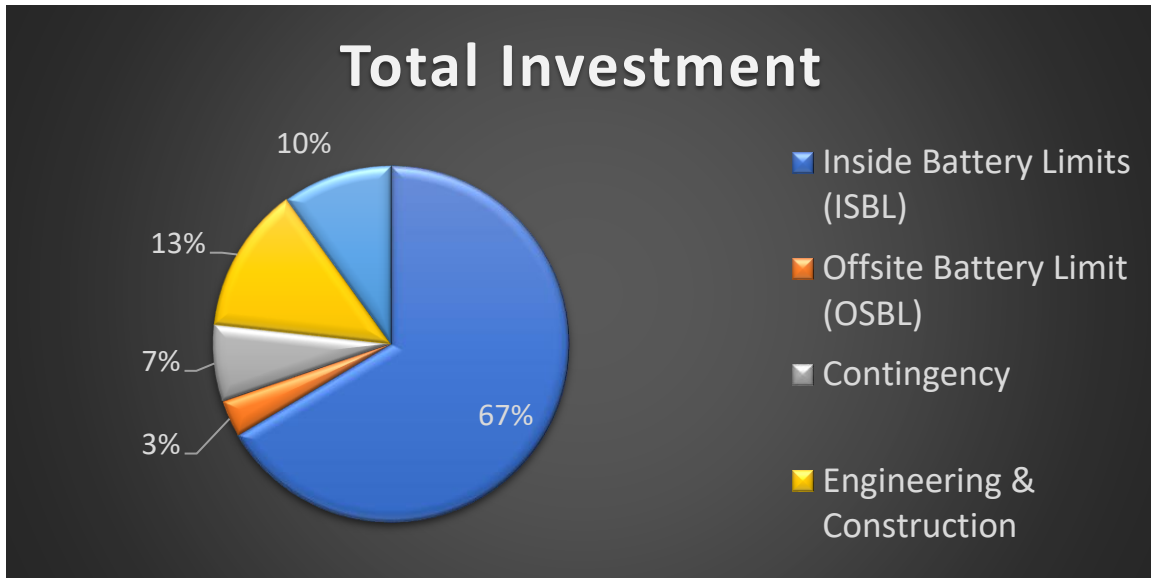


Figure 14: Total Investment Breakdown

The Net Present value of this company was analyzed with three different MACRS depreciation time span, 5, 7, and 15 years. Figure 14 below displays the side by side comparisons of the NPV and the rate of return (DCFRROR/IRR). The graph illustrates that the 5-year recovery supplies a higher NPV and DCFRROR with a \$325.7 MM and 42.4% respectively. Figure 15 illustrates the NPV over the 20 year lifespan of the treatment, The horizontal axis go from 1 to 21 since the 1 is technically year 0 in which half the total investment is paid and the other half is paid the first year of operating. The fourth year is when profit commences since the treatment is operating its maximum. From year 4 and moving forward the NPV increases each year. The tax rate is 20% based on the 2018 Tax Reduction Act, the mean discount rate is 8%, and the depreciation goes from year 2 to 7, the whole NPV table can be found in Appendix.

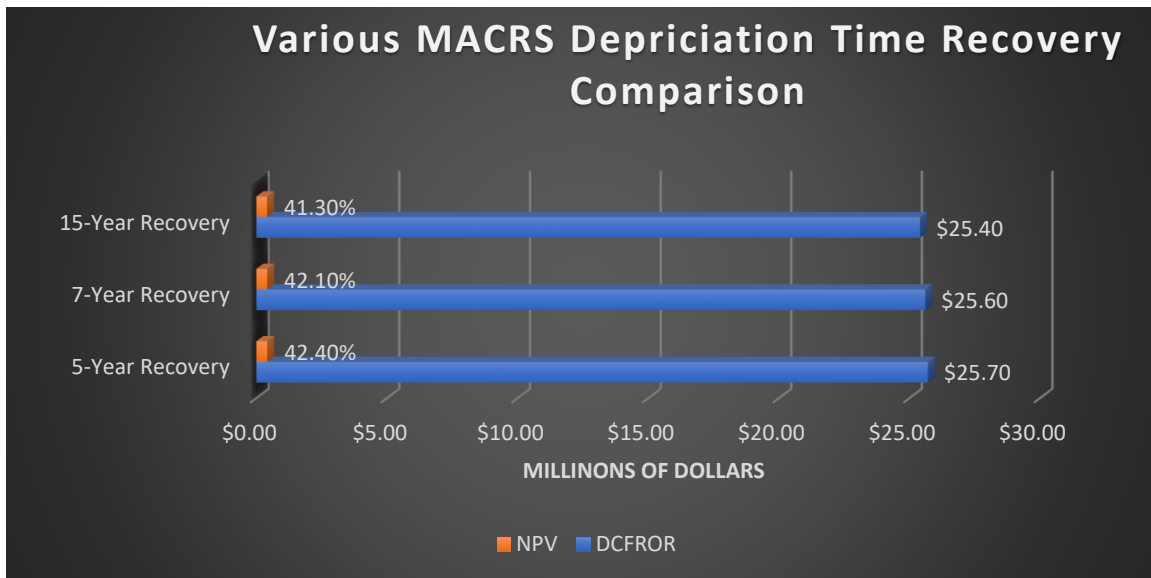


Figure 15: Varying MACRS Depreciation Time Recovery Comparisons

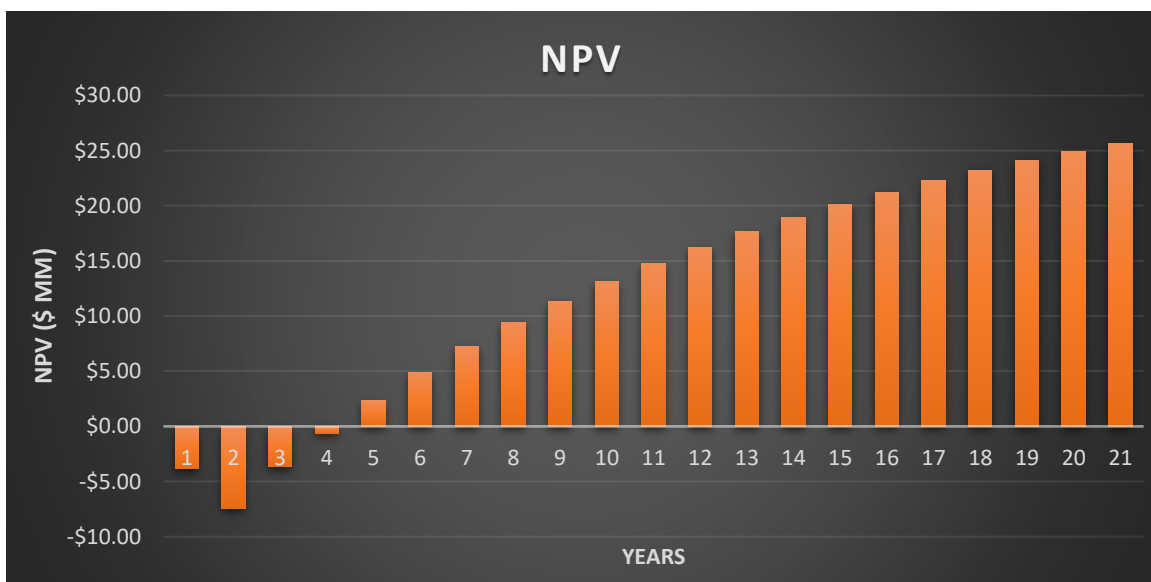


Figure 16: Net Present Value Over the Lifetime

From the NPV MACRS Depreciation comparison, the 5-year recovery was the one utilized when doing a sensitivity analysis for the company. For the purpose of analyzing how the NPV

would be affected by varying parameters the following six were selected to be analyzed: raw material unit price, product unit price, utilities which in this case was only diesel, FCOP, ISBL and total capital cost. The sensitivity analysis was based on the size of the project and by size it would mean staying at site longer and using a larger lagoon to store the excess flow or by increasing the number of trailers at the site to match the capacity. In other would varying size would be based on numbering up or down to meet whatever the demand is. The higher percent change from the base line, the higher or lower the NPV. When looking at the size of the plant, the product line has the most effect as expected. The higher the size the lower the NPV but higher the product the higher the NPV. The FCOP line is the one matching the size more closely than the other lines since both the size and fixed cost do not change much since they are fixed prices. The diesel utility cost is the most linear since the cost to the overall process is small compared to other variables therefore a somewhat linear trend surface. ISBL and the diesel cost have a downward linear trend since the bigger the treatment size the more equipment, installation, and fuel that is needed to meet operation requirements, therefore the NPV would go down with their increase. Figure 16 below depicts the analyzed parameters and their effects side by side.

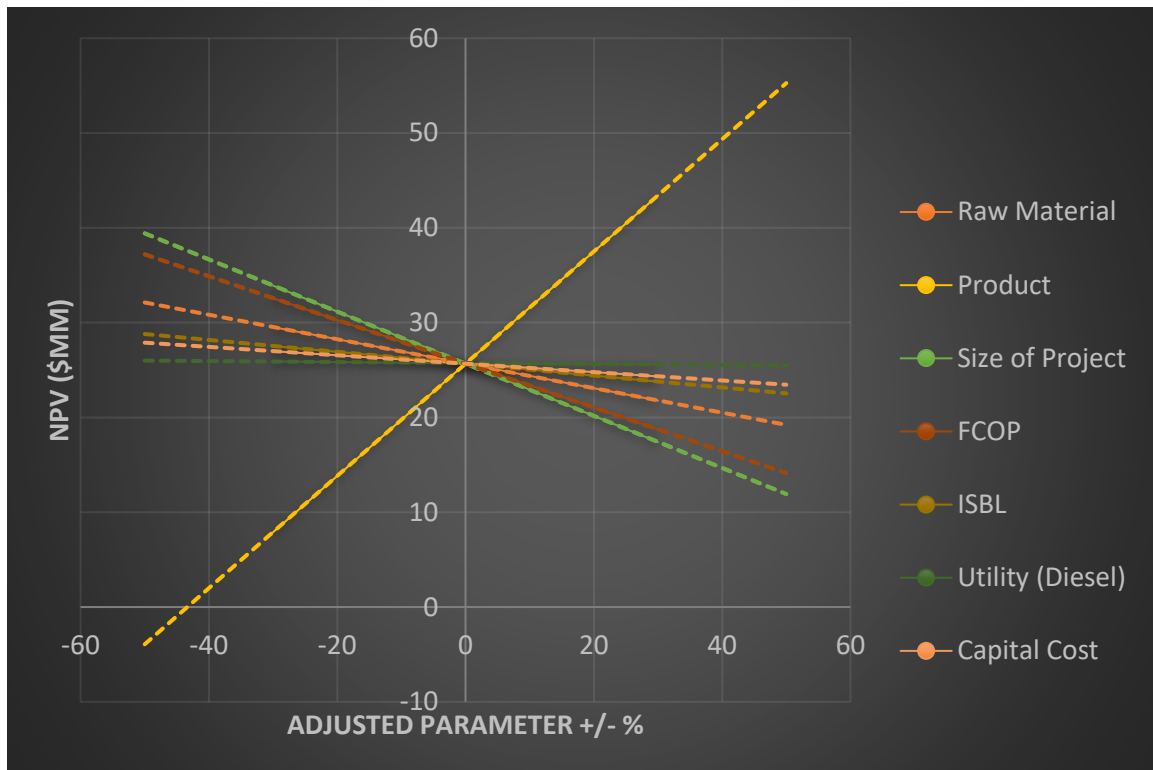


Figure 17: Sensitivity Analysis

Figure 18 below depicts the sensitivity of product price as a function of project size. This shows how in order to be profitable the pricing of the service needs to be higher as the project gets smaller. Both the product cost and sizing of treatment are the most sensitive factors in this project so the ideal choice would be one that balances both, while still achieving a high net present value. There are other variables that impact the profit almost as close as the Size. These are the FCOP and ISBL which have the similar trend as the sizing since they are correlating with how big the project is. As more water needs to be treated, the sized increases which means equipment increases (ISBL) as well as the components that make operations run. Overall, numbering up or down based on the flowback water quantity would not be an issue since the sizing is inversely related to the rest of the studied factors in the sensitivity analysis.

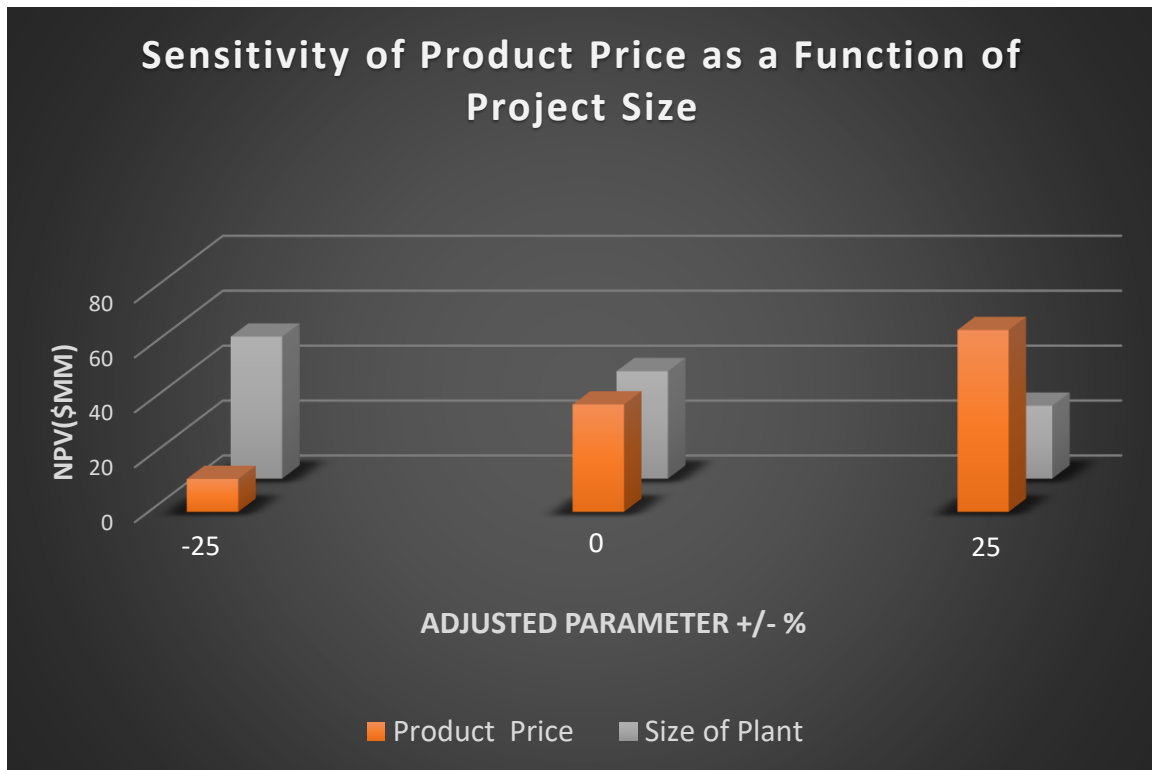


Figure 18: Sensitivity of Product Price as a Function of Project Size

The summary of the whole economic analysis is in the table below. The revenue is \$12.6MM, total investment \$7.71MM, annual cash flow is \$4.40MM the capital cost is \$8.21 which is the sum of the FCOP and VCOP. The ROI is 57% in 3.80 years and the gross profit is \$9.36MM. Overall, this treatment is a very profitable investment as seen by the NPV and DCFROR being \$25.70MM and 42.40% respectively. The higher the DCFROR the be better the project since it has the ability to profit even with higher depreciation rates.

Table 12: Summary of the Economic Analysis

FCOP:	\$4.83MM
VCOP:	\$3.38MM
REVENUE:	\$12.6MM
TOTAL INVESTMENT:	\$7.71MM
CCOP:	\$8.21MM
AVG ANNUAL CASH FLOW:	\$4.40MM
SIMPLE PAYBACK TIME (YRS):	1.75
ROI (%):	56.98%
ROI (YRS):	3.80
GROSS PROFIT:	\$9.36MM

Recommendation

If an investor were to move forward to bring this project to life there are a few recommendations to ensure the mobile treatment is a successful and profitable business. First, modeling of various ways that equipment may be integrated in one truck should be modeled to fit on fewer trucks and reduce the costs needs to be. This can also help to ensure that when numbering up or down, the correct number of trucks are dispatched to the fracking sites. Secondly, more information about activated carbon should be found for its capacity of absorbing components. Contacting vendors and exploring the option of regenerating the carbon on site could also improve the overall economics. Third, the membrane and filtration module have the potential to be optimized to reduce the power consumed and membrane fouling. Another point to look at is finding the exact sites that the waste produced such as oil and extra unwanted components produced should

be transported to for disposal. Lastly, exploring more potential applications of mobile treatment may bring further revenue such as dealing with accidental leaks from a chemical plant.

Appendix

Design Basis

Our design basis is located at one of the largest shale formations in the U.S., the Marcellus shale. The only raw material that will be required in our mobile solution is the activated carbon that is used in absorbing organic materials. The dosage of carbon that is essential in removing unwanted material efficiently is assumed to be able to achieve 10 percent loading. As for our main feed of water, the volume varies by region. A disclosure in West Virginia reported a mean of approximately 5 million gallons of water [26]. As mentioned earlier, a majority of fracking water is lost underground, thus we will assume about 30% of that water will return as our flowback feed stream. In addition, a typical fracking site takes about 3-5 days in fracking rock formation [27]. Thus, we will assume it will take 3 days to frack, so dividing 3 days by the total volume of feed water leads to about 500 thousand gallons a day of flowback water. It was reported that hydraulic fluid consists of 90% fresh water, 9% sand, and 0.5% in chemical additives [28]. Therefore, those details will remain consistent with the 30% of flowback water that we are assuming to retrieve back, in addition to the other materials that come from underground. The following table represents the concentrations of flowback water from a Marcellus shale disclosure.

Table 13: Flowback Water Concentration from Marcellus Shale [29]

Cations, metals and metalloids			Organics			Anions		
<i>Sample</i>	<i>Units</i>	<i>Value</i>	<i>Sample</i>	<i>Units</i>	<i>Value</i>	<i>Sample</i>	<i>Units</i>	<i>Value</i>
Ba	mg/L	514.68	Benzene	µg/L	194.47	Nitrate	mg/L	0.02
Sr	mg/L	1365.00	Ethylbenzene	µg/L	85.34	Nitrite	mg/L	0.06
Na	mg/L	26202.00	Styrene	µg/L	141.00	SO4	mg/L	55.93
Mg	mg/L	835.00	Toluene	µg/L	621.71	Cl	mg/L	42683.00
Ca	mg/L	7269.00	Xylene	µg/L	825.75	PO4	mg/L	9.49
K	mg/L	260.66	Ethane	µg/L	815.99	Br	mg/L	465.96
Fe	mg/L	67.08	Methane	µg/L	3420.00			
Mn	mg/L	5.50	Propane	µg/L	1130.00			
Cr	mg/L	0.08	Diesel	mg/L	60.56			
Pb	mg/L	0.10	Gas	mg/L	25.75			
Se	mg/L	0.26	Oil	mg/L	13.27			
Al	mg/L	4.61						
Zn	mg/L	0.14						

Block Flow Diagram (BFD)

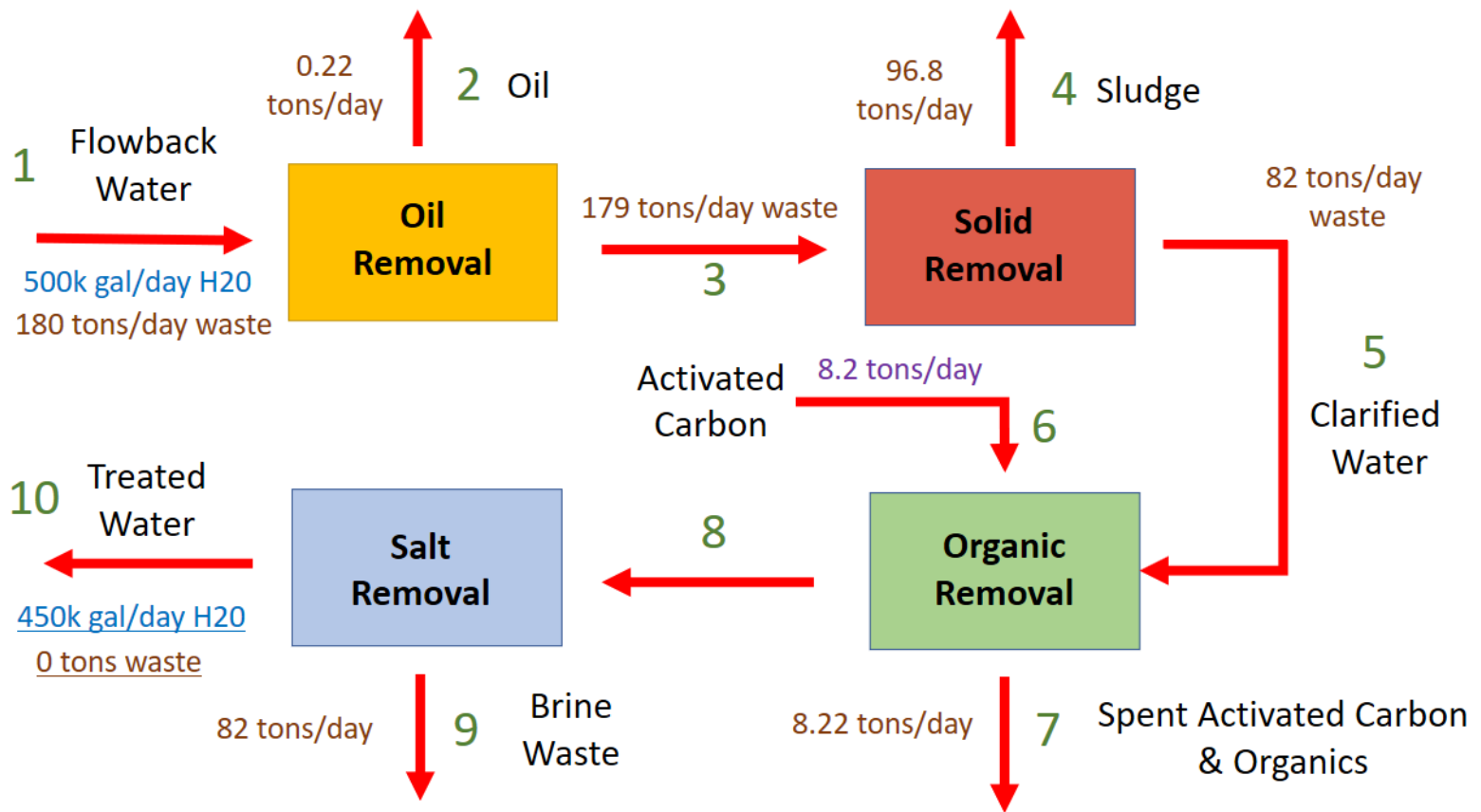


Figure 19: Block Flow Diagram of Flowback Treatment

Process Flow Diagram (PFD)

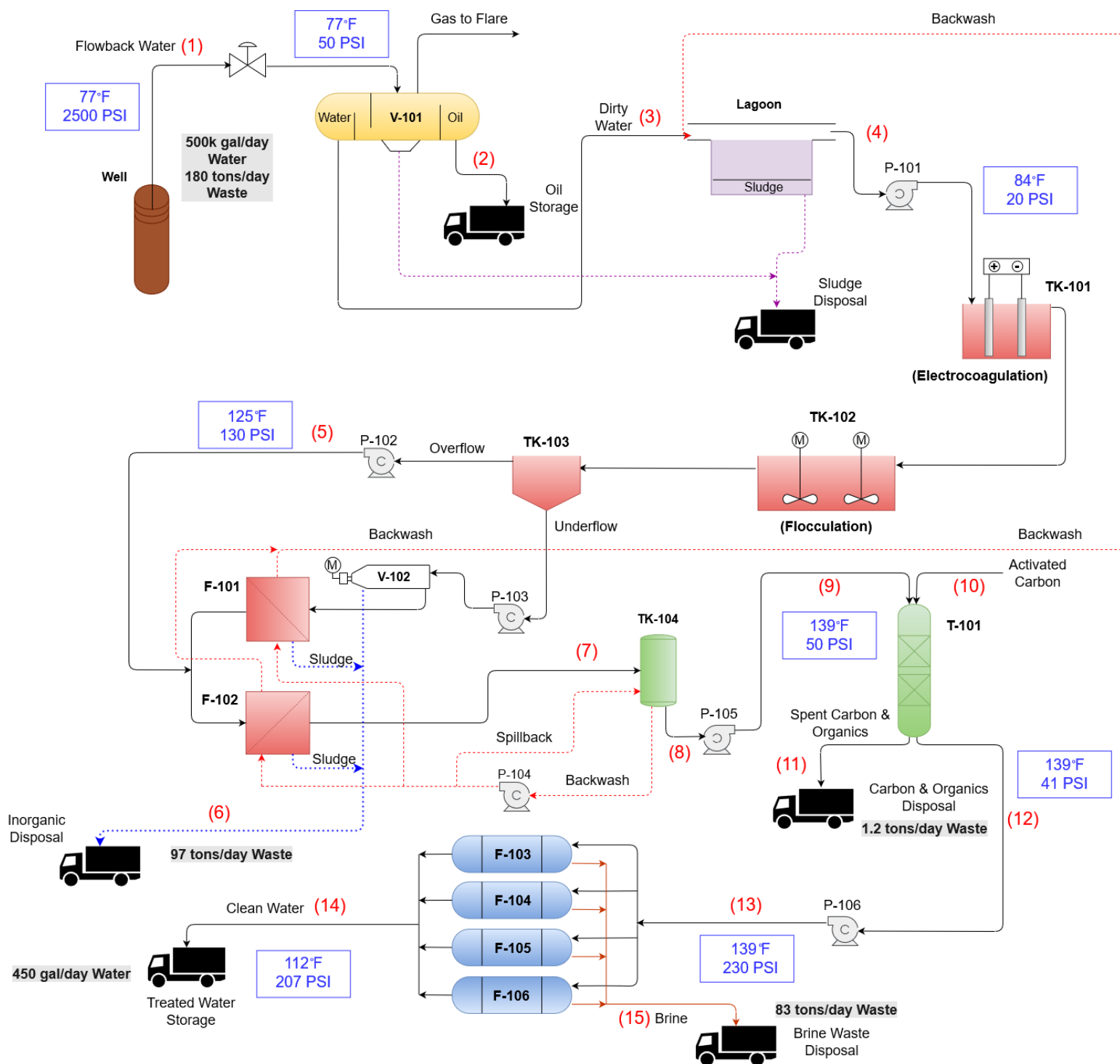


Figure 20: Process Flow Diagram for Flowback Treatment

Equipment & Process Descriptions

1. **API Separator (V-101):** Flowback water flows back from the fractured well at a high pressure, through a pressure-reducing valve to the API Separator. This separator is where any oil is separated from the water by gravity, in which the baffles are sized based on the density of each fluid. Any hydrocarbons are flared from the top, then the contaminated water flows to the lagoon.

2. **Electrocoagulation (TK-101):** As the water settles in the lagoon, it enters the electrocoagulation tank. This is where an electric current is sent into the tank with an anode and cathode plate. As electricity is sent into the system, contaminated particles begin to come together due to changing the particles surface charge.

3. **Flocculation (TK-102):** After electrocoagulation, the solution is sent to the flocculation tank where it will be slowly mixed to create fine particulates to clump together.

4. **Settler (TK-103):** Once the contaminants begin to ‘floc’ together, it enters the settler tank where particles are separated from liquids by gravity. Two streams exit the settler. The top, supernatant stream enters the second stage of microfiltration, while the bottom streams enters a decanter to further remove additional sediment prior to filtration.

5. **Decanter Centrifuge (V-102):** The bottom stream from the settler enters the decanter centrifuge. This is where further solids are removed from the contaminated water by the slurry entering a bowl that is rotating at high speeds. Sludge will be pressed outwards from the wall and pumped back to the lagoon, while the water will go through filtration.

6. **Submerged Microfiltration (F-101/2):** Water from the settler will go through 2 stages of microfiltration, depending on whether it’s coming from the bottom or top of the settler. Microfiltration used to separate solids from liquids by using a filter medium, such as a membrane,

to limit the particle diameter of the solids that can pass through the membrane's pores that's about 0.1 micron.

7. **Backwash Drum (TK-104):** This backwash drum is used to gain a supply of effluent post microfiltration to be able to clean out any accumulated particles in the membranes by having a reverse flow. Backwash stream will then lead back to the lagoon to be further treated.

8. **Activated Carbon Adsorber (T-101):** Effluent from microfiltration enters the top of the adsorber column. This is where the activated carbon will be used as the absorbent in a separating organic component from the water.

9. **Reverse Osmosis (F-103):** Exiting water from the activated carbon adsorber is then pumped to a high pressure of 60 psi to reverse osmosis. A high pressure is needed to force molecules through the membrane to remove any leftover impurities from water. Leftover brine gets collected, while the clean water is sent to a truck for reuse.

10. **Pump (P-101-6):** a device used in moving fluids mechanically leading to an increase of pressure.

Equipment Sizing & Materials

Table 14: Complete Equipment Summary [49-53]

Unit ID	Type	Size/Power	Material	Short Description
V-101	API Separator	Diameter: 7.5 feet Length: 36.3 feet	Carbon Steel	A gravity separation device used to separate oil from water. Device uses gravity in leading to a rise of oil particles based on its density.
TK-101	Electrocoagulation Tank (Rapid Mixing)	Diameter: 6.5 feet Height: 7 feet	The electrodes are designed as plates and they are composed of iron and aluminum. The tank itself is made of steel and concrete.	Electrical current is used that coagulates organic constituents and suspended solids in water. It has three major parts: reaction chamber (contains multiple anode and cathode pairs in which the water passes through) an electrical current; And a system to dewater the precipitated/coagulated solids
TK-102	Flocculation (Slow Mixing Tank)	Diameter: 6.5 feet Height: 7 feet	FRP (Fiber Reinforced Plastic) for the Tank and Steel frame	Slow mixing process where particles are brought into contact to promote agglomeration (mass collection). This increases the number of contacts between the particles from prolonged agitation.
TK-103	Sedimentation Tank	Diameter: 5 feet Length: 11.8 ft	FRP (Fiber Reinforced Plastic) for the Tank and Steel frame	A sedimentation tank allows suspended particles to settle out of water or wastewater as it flows slowly through the tank, thereby providing some degree of purification. The retention period is in between 4 hours to 8 hours.
V-102	Decanter Centrifuge	Diameter: 3.5 feet Capacity: 10 CF	316SS: Chromium-nickel- molybdenum austenitic stainless steel developed to provide improved corrosion resistance	Separates solids from liquids using centrifugal force and has a horizontal orientation.
F-101-2	Microfiltration - Submerged	Frame Capacity: 30 CF	Organic based material: polymer-based membrane and the membrane housing is FRP	Submerged RO system allows crossflow in microfiltration systems in order to remove particles more efficiently with backwash to lessen accumulation

TK-104	Backwash Storage Tank	Diameter: 5 feet Length: 12.1 feet	FRP	A storage tank to retain enough clean water to provide backwash to regenerate the microfilters. Backwashing consists of reversing the flow of water so that it enters from the outlet side of the filter to dislodge the solid layer from the filter media.
T-101	Carbon Adsorption Packed Column	ASPEN Height: 25 feet	FRP	A device used to remove minor components from a solution by adsorbing them onto an adsorbent, activated charcoal in this case.
		Hand Calculated: 20 feet and Diameter: 2 feet		
		Thickness: 0.016feet		
F-103	Spiral Wound Reverse Osmosis	Tank Size: 30,000-gal Height: 24.3 feet Diameter: 14.5 feet	Membrane Housing is FRP. Membrane is 3 layers: Polyester supports the web, a microporous polysulfide interlayer and an ultra-thin polyamide barrier layer on the top surface	A process that is used to remove impurities from water by using pressure to force molecules through a membrane.

Truck Arrangement

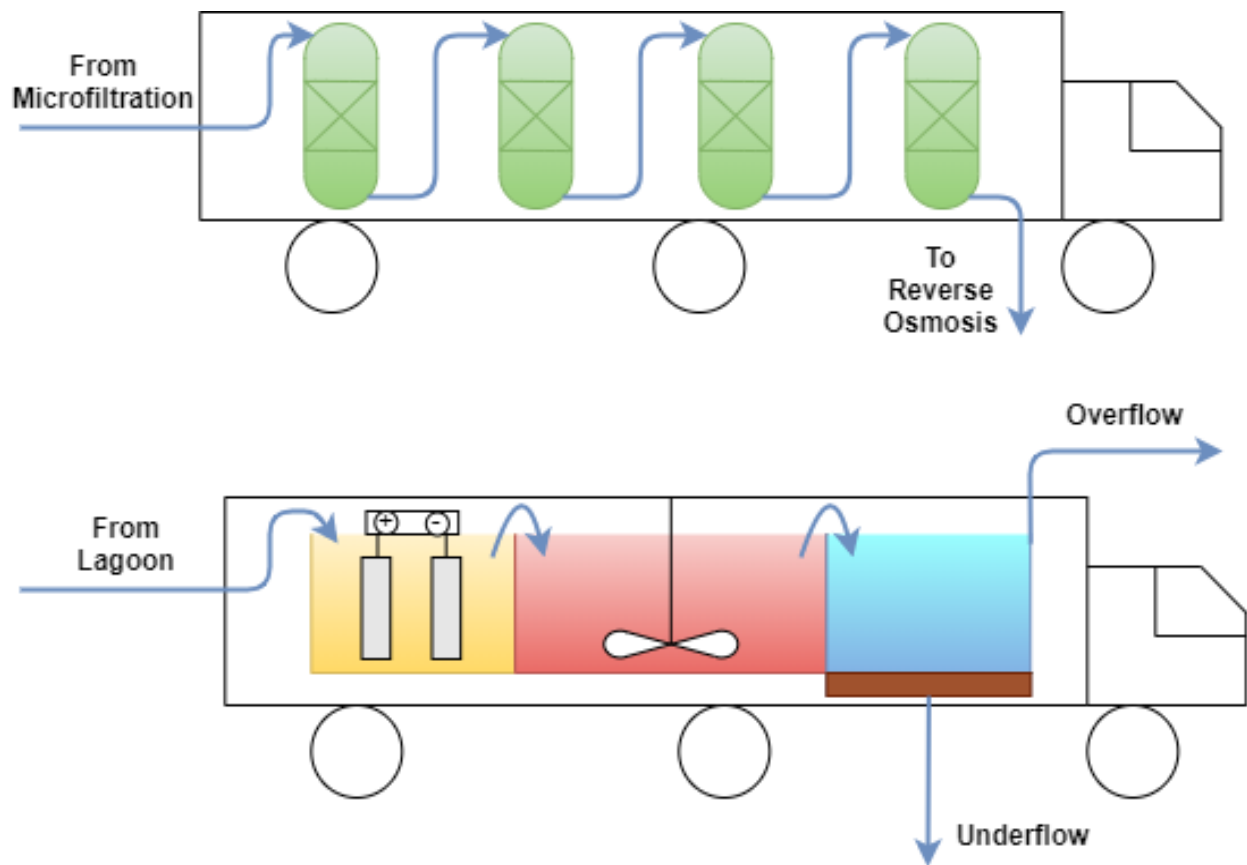


Figure 21: Truck Arrangement

Aspen Simulation

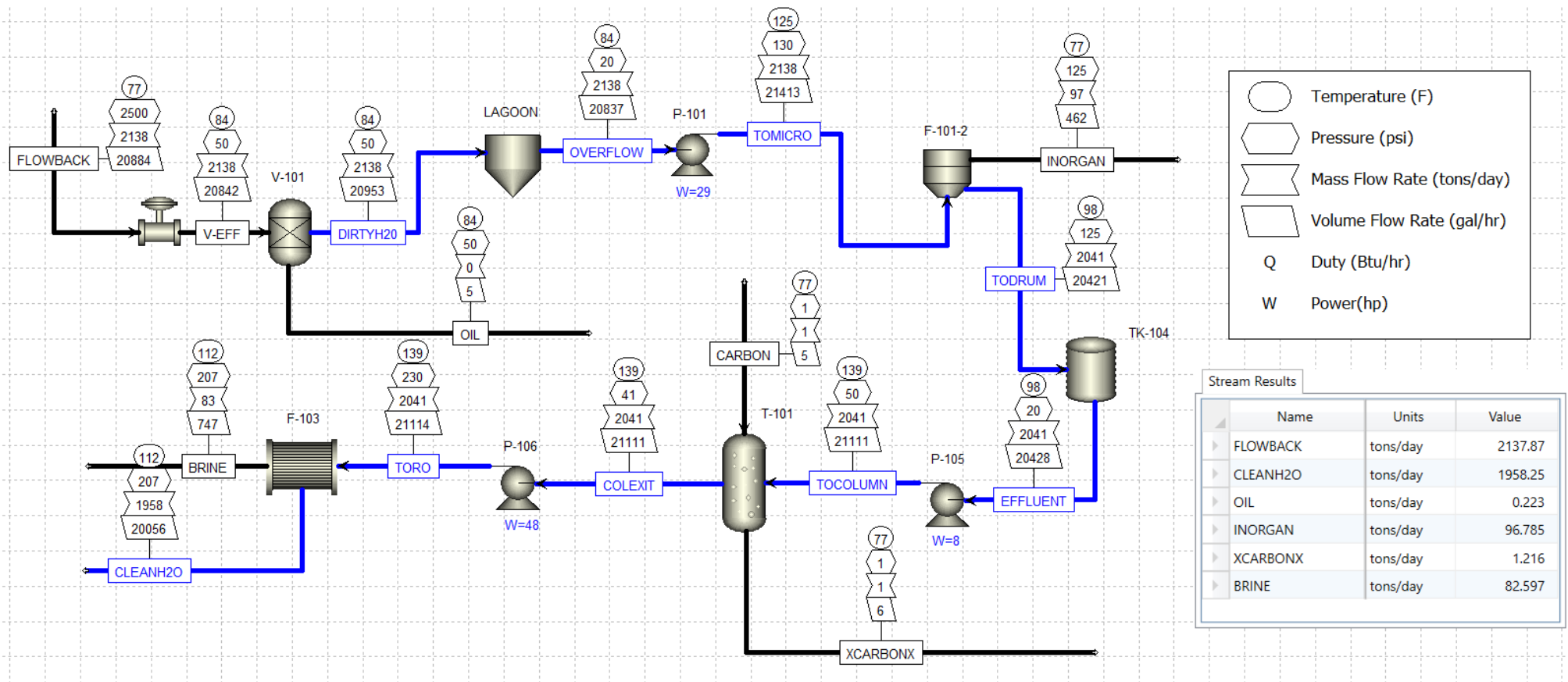


Figure 22: Aspen Simulation Results

Material Balance

Assumptions for Mass Balance

- 1) Estimated the amount of flowback water was 30% volume from initial fracking fluid.
- 2) Separation yield of 100% solid removal throughout process.
- 3) The amount of coagulate and absorbent was 30% of the total mass of components.
- 4) V-101 (API Separator) is a ‘SEP’ model, a perfect split of undissolved oil from contaminated water.
- 5) Carbon stream treated as continuous stream; Batch process cannot be simulated with ASPEN. Carbon quantity estimated assuming water is saturated with hydrocarbon at ambient temperature, 60°F, and 1 week between adsorbent reloads at design flow rate.

Table 15: Material Balance – Part 1

Stream Parameter	Units	FLOWBACK	OIL	DIRTYH2O	OVERFLOW	TOMICRO
Stream # on PFD		1	2	3	4	5
From			V-101	V-101	LAGOON	P-101
To		V-101		LAGOON	P-101	F-101-2
Temperature	F	77.00	77.00	77.00	83.62	124.61
Pressure	psi	2500.00	2500.00	2500.00	20.00	130.00
Mass Liquid Fraction		0.92	1.00	0.92	0.92	0.92
Mass Enthalpy	Btu/lb	-6269.92	-10156.79	-6245.59	-6245.59	-6245.18
Mass Density	lb/cuft	63.81	26.95	63.82	63.95	62.24
Average MW		18.87	3.53	18.88	18.88	18.88
Volume Flow	gal/hr	20884.32	5.16	20879.16	20835.97	21409.64
Mass Flows	tons/day	2137.87	0.22	2137.65	2137.65	2137.65
IRON	tons/day	0.15	0.00	0.15	0.15	0.15
LEAD	tons/day	0.00	0.00	0.00	0.00	0.00
ZINC	tons/day	3.06	0.00	3.06	3.06	3.06
OIL	tons/day	0.22	0.22	0.00	0.00	0.00
WATER	tons/day	1959.25	0.00	1959.25	1959.25	1959.25
BARIUM	tons/day	1.15	0.00	1.15	1.15	1.15
SODIUM	tons/day	58.67	0.00	58.67	58.67	58.67
MAGNESIUM	tons/day	1.87	0.00	1.87	1.87	1.87
CALCIUM	tons/day	16.28	0.00	16.28	16.28	16.28
POTASSIUM	tons/day	0.58	0.00	0.58	0.58	0.58
MANGANESE	tons/day	0.01	0.00	0.01	0.01	0.01
CHROMIUM	tons/day	0.00	0.00	0.00	0.00	0.00
ALUMINUM	tons/day	0.01	0.00	0.01	0.01	0.01
BENZENE	tons/day	0.00	0.00	0.00	0.00	0.00
ETHYLBENZENE	tons/day	0.00	0.00	0.00	0.00	0.00
STYRENE	tons/day	0.00	0.00	0.00	0.00	0.00
TOLUENE	tons/day	0.00	0.00	0.00	0.00	0.00
ETHANE	tons/day	0.00	0.00	0.00	0.00	0.00
METHANE	tons/day	0.01	0.00	0.01	0.01	0.01
PROPANE	tons/day	0.00	0.00	0.00	0.00	0.00
CHRLORINE	tons/day	95.56	0.00	95.56	95.56	95.56
BROMINE	tons/day	1.04	0.00	1.04	1.04	1.04
CARBON	tons/day	0.00	0.00	0.00	0.00	0.00

Table 16: Material Balance – Part 2

Stream Parameter	Units	INORGAN	TODRUM	EFFLUENT	TOCOLUMN	CARBON
Stream # on PFD		6	7	8	9	10
From		F-101-2	F-101-2	TK-104	P-105	
To			TK-104	P-105	T-101	T-101
Temperature	F	77.00	98.00	98.28	138.52	77.00
Pressure	psi	125.00	125.00	20.00	50.00	1.00
Mass Liquid Fraction		0.00	0.96	0.96	0.96	0.00
Mass Enthalpy	Btu/lb	-12.66	-6527.10	-6527.10	-6526.98	0.00
Mass Density	lb/cuft	130.62	62.30	62.28	60.26	140.46
Average MW		70.61	18.25	18.25	18.25	12.01
Volume Flow	gal/hr	461.89	20420.64	20428.07	21111.30	5.33
Mass Flows	tons/day	96.78	2040.86	2040.86	2040.86	1.20
IRON	tons/day	0.00	0.15	0.15	0.15	0.00
LEAD	tons/day	0.00	0.00	0.00	0.00	0.00
ZINC	tons/day	0.00	3.06	3.06	3.06	0.00
OIL	tons/day	0.00	0.00	0.00	0.00	0.00
WATER	tons/day	0.18	1959.07	1959.07	1959.07	0.00
BARIUM	tons/day	0.00	1.15	1.15	1.15	0.00
SODIUM	tons/day	0.00	58.67	58.67	58.67	0.00
MAGNESIUM	tons/day	0.00	1.87	1.87	1.87	0.00
CALCIUM	tons/day	0.00	16.28	16.28	16.28	0.00
POTASSIUM	tons/day	0.00	0.58	0.58	0.58	0.00
MANGANESE	tons/day	0.00	0.01	0.01	0.01	0.00
CHROMIUM	tons/day	0.00	0.00	0.00	0.00	0.00
ALUMINUM	tons/day	0.00	0.01	0.01	0.01	0.00
BENZENE	tons/day	0.00	0.00	0.00	0.00	0.00
ETHYLBENZENE	tons/day	0.00	0.00	0.00	0.00	0.00
STYRENE	tons/day	0.00	0.00	0.00	0.00	0.00
TOLUENE	tons/day	0.00	0.00	0.00	0.00	0.00
ETHANE	tons/day	0.00	0.00	0.00	0.00	0.00
METHANE	tons/day	0.00	0.01	0.01	0.01	0.00
PROPANE	tons/day	0.00	0.00	0.00	0.00	0.00
CHRLORINE	tons/day	95.56	0.00	0.00	0.00	0.00
BROMINE	tons/day	1.04	0.00	0.00	0.00	0.00
CARBON	tons/day	0.00	0.00	0.00	0.00	1.20

Table 17: Material Balance – Part 3

Stream Parameter	Units	XCARBONX	COLEXIT	TORO	BRINE	CLEANH2O
Stream # on PFD		11	12	13	14	15
From		T-101	T-101	P-106	F-103	F-103
To			P-106	F-103		
Temperature	F	77.00	138.52	138.54	77.00	77.00
Pressure	psi	1.00	41.00	60.00	54.00	54.00
Mass Liquid Fraction		0.00	0.96	0.96	0.01	1.00
Mass Enthalpy	Btu/lb	0.00	-6527.06	-6526.99	-67.99	-6866.65
Mass Density	lb/cuft	133.83	60.26	60.26	69.22	62.05
Average MW		12.10	18.25	18.25	26.19	18.02
Volume Flow	gal/hr	5.66	21110.94	21111.22	743.88	19673.06
Mass Flows	tons/day	1.22	2040.85	2040.85	82.60	1958.25
IRON	tons/day	0.00	0.15	0.15	0.15	0.00
LEAD	tons/day	0.00	0.00	0.00	0.00	0.00
ZINC	tons/day	0.00	3.06	3.06	3.06	0.00
OIL	tons/day	0.00	0.00	0.00	0.00	0.00
WATER	tons/day	0.00	1959.07	1959.07	0.82	1958.25
BARIUM	tons/day	0.00	1.15	1.15	1.15	0.00
SODIUM	tons/day	0.00	58.67	58.67	58.67	0.00
MAGNESIUM	tons/day	0.00	1.87	1.87	1.87	0.00
CALCIUM	tons/day	0.00	16.28	16.28	16.28	0.00
POTASSIUM	tons/day	0.00	0.58	0.58	0.58	0.00
MANGANESE	tons/day	0.00	0.01	0.01	0.01	0.00
CHROMIUM	tons/day	0.00	0.00	0.00	0.00	0.00
ALUMINUM	tons/day	0.00	0.01	0.01	0.01	0.00
BENZENE	tons/day	0.00	0.00	0.00	0.00	0.00
ETHYLBENZENE	tons/day	0.00	0.00	0.00	0.00	0.00
STYRENE	tons/day	0.00	0.00	0.00	0.00	0.00
TOLUENE	tons/day	0.00	0.00	0.00	0.00	0.00
ETHANE	tons/day	0.00	0.00	0.00	0.00	0.00
METHANE	tons/day	0.01	0.00	0.00	0.00	0.00
PROPANE	tons/day	0.00	0.00	0.00	0.00	0.00
CHRLORINE	tons/day	0.00	0.00	0.00	0.00	0.00
BROMINE	tons/day	0.00	0.00	0.00	0.00	0.00
CARBON	tons/day	1.20	0.00	0.00	0.00	0.00

Sample Calculation – Adsorption Column [30]

*TYPE OF CARBON

Bituminous GAC is the one most frequently used for treating low concentrations of low molecular weight organic contaminants in the aqueous phase. Bituminous coal will also have a more fully developed pore distribution, including “transport pores” that improve the rate of adsorption making it effective for water treatment. Bituminous GAC has a large surface area, ~900 m²/g, and an apparent density of ~0.50 g/cm³ (30 lb/ft³). These carbons are usually harder than others so they are more abrasion resistant, and can be more vigorously backwashed without damage.

*PRESSURE DROP

Typical loading rates are 2–6 gpm/ft² sometimes, loadings up to 10 gpm/ft² are used. Loadings greater than 6 gpm/ft² typically result in bigger head losses and big pressure drops. For this case the pressure difference is 9psi and we can assume loading rate of 4gpm/ft²

*The figure below displays the breakthrough curve for the carbon and for the purposes of this project the assumption is that the breakthrough will occur $\frac{3}{4}$ of the capacity. The assumption is made because having it as $\frac{3}{4}$ it helps to treat the most volume. Because the adsorption process is exothermic, increases in temperature may reduce the degree of adsorption so this temperature effect is not accounted in water treatment applications.

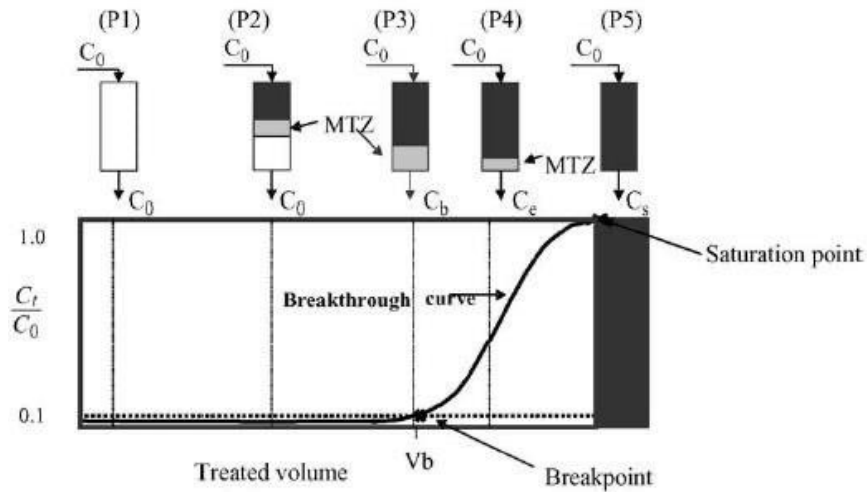


Figure 23: Breakthrough Curve for Adsorption

*OPERATION CONDITIONS

*CONTACT TIME

To go from a medium ppm range to a low ppb range requires approximately 30 minutes EBCT (empty bed contact time).

$$EBCT = V/Q \text{ or } LA/Q$$

V = bulk volume of GAC in contactor, m³ (ft³)

A = cross-sectional bed area, m² (ft²)

L = bed depth, m (ft)

Q = volumetric flow rate, L/s (ft³/min or gal/hr)

$$EBCT = \frac{V}{Q} \text{ } 0.5 \text{ hr} = \frac{V}{\frac{21111.40 \text{ gal}}{\text{hr}}}$$

Volume=10555.65 gal

***VOLUME OF THE ADSORBED**

The vessel should have an additional 20–50% bed expansion for backwashing the carbon before placing in service. This expansion allowance is critical in systems where suspended solids are expected, or there is no pre-filtration. For this case the estimated expansion will be assumed to be 40%, The adsorber volume is then calculated from:

$$V = \frac{CUR * COR}{DENSITY} * SF$$

V=volume of adsorbed

CUR= carbon usage rate, g/day (lb/day)

COP= carbon change out period, days

ρ = bulk density of carbon, g/cm³ (lb/ft³)

S.F. = safety factor to provide extra non-carbon-containing volume for operational uncertainty, 1.2–1.5.

*Carbon replacement rate is 379lbs of carbon per hour

$$V = \frac{171912g}{\frac{0.5g}{c \ m^3}} * 1.4 = 4813.53m^3$$

***BED DEPTH L & DIAMETER**

direct function of the contactor diameter and volume. Bed depth (L) can be solved knowing the adsorber volume (V) and adsorber bed area (A) using the equation:

$$L = \frac{V}{A} = \frac{4813.53m^3}{800m^3} = 6.01m = 19.72feet = \sim 20feet$$

The ratio of L/D is about 10 therefore the diameter = 2feet

***CARBON USAGE**

Estimate GAC usage is based on data using the relationship for a batch system: Relationship 3:1

is generally used to estimate carbon usage for batch systems

$$CUR = \frac{Co - CeF}{\left(\frac{x}{m}\right)_{Co}} = 50,000lbs \text{ per } 16 \text{ days is the estimate}$$

Co = initial concentration (mg/L)

C = desired effluent concentration mg/L (mg adsorbed/L carbon) (x/m)_Co= mg contaminated/L carbon

CUR = carbon usage rate (g/day)

F = volumetric flow rate of contaminated liquid treated/day (L/day)

COMPARISON OF ASPEN AND HAND CALCULATIONS

The hand calculation required multiple aspects to be considered such as the pressure drop, contact time, type of carbon, carbon usage, breakpoint curve, and the length/depth ratio. There were various assumptions made since data was lacking to correctly determine the breakthrough curve. Also the carbon information was taken from averages as well as the ratio of length to diameter. The pressure drop was taken to be a common one as well. All these assumptions and averages cause for possible inaccurate results. ASPEN on the other hand has more options and takes the actual data from the simulation to size the equipment. All the equipment sizing was

extracted from the simulation and only the adsorber was done by hand and used to compare both results here.

When observing the absorber column, it can be noted that the height obtained from ASPEN was 5 feet taller than the hand calculations. Also, the diameter was 3 feet longer than the hand calculation, which is 150% higher. When looking at the overall percent difference the ASPEN results do have a slightly higher sizing of equipment by an average of 87.5%. The reason being that there were more details that can be included in the ASPEN software which is one reason these results occur.

Utility Usage & Calculation

Table 18: Electricity Usage

Equipment Name	Type	Rate	Units	Operating condition
P-101	Electricity	5	hp	Increasing the pressure of 20836 gal/hr of water from 0 psi to 20 psi
P-102	Electricity	29	hp	Increasing the pressure of 20836 gal/hr of water from 20 psi to 130 psi
P-103	Electricity	29	hp	Increasing the pressure of 20836 gal/hr of water from 20 psi to 130 psi
P-104	Electricity	5	hp	Increasing the pressure of water from 0 psi to 20 psi
P-105	Electricity	8	hp	Increasing the pressure of 20428 gal/hr of water from 20 psi to 50 psi
P-106	Electricity	48	hp	Increasing the pressure for Reverse osmosis membrane
Electrocoagulation	Electricity	3	kW	Oxidizing the anode made of iron and aluminum
Motor for Slow Mixing Tank	Electricity	89	kW	Stirring the 37 CF liquid to settle the flocculating solids

Decanter Centrifuges	Electricity	115	hp	Separating solids from liquids
Total		270.23	kW	All the equipment's work at ambient temperature

For this project, one of the major sources of utility that is needed is electricity. Due to our process being a mobile water treatment, a generator will be bought in order to supply the electrical needs. The utility of each pump was estimated by Aspen.

The utility of electrocoagulation and decanter centrifuges was estimated based on the power of existing commercial equipment. The calculation of the power of Decanter Centrifuges based on the type RS-1800RS from Rotating Solutions Inc. It has a 75 hp main drive motor and a 40 hp back drive motor, which can treat 350 GPM water [30].

Electrocoagulation

The power is based on the type QLW-100 from *qinglan*. It has a capacity of 26417.2 gal/hr and its power is 3 kW [31].

Flocculation

The electricity of the motor for slow mixing tank was calculated as follows:

$$(Re)_i = N_i * D_i^2 * \frac{Density}{Viscosity}$$

Where the $(Re)_i$ is the impeller Reynolds number. N_i is the stirred speed which is 2 s^{-1} .

D_i is the diameter of propeller, here using a marine propeller 2 m in diameter. Density is $1000 \frac{\text{kg}}{\text{m}^3}$.

Viscosity at $1 \times 10^{-2} \text{ Pa}\cdot\text{s}$. The result is $(Re)_i = 8 \times 10^5$ Flow at this $(Re)_i$ is fully turbulent,

according to the power number and Reynolds number curve [32], the power number $N_p = 0.35$.

Therefore: $P = N_p' * Density * N_i^3 * D_i^5 = 89 kW$

Diesel Fuel for Generator [46]

The generator that is bought in order to deliver electricity for the pumps and motors will use approximately 16 gallons per hour. As our treatment is planned to last 3 days, we are assuming the generator will be on for 24 hours. These characteristics will lead to about 1,159 gallons of diesel per site treated.

Economic Analysis Including Equipment Costs

Table 19: ISBL & Installation Costs

	Equipment	Quantity	ASPEN COST \$2019
1	API Separator (V-101)	1	\$ 97,800
2	TK-101	1	\$ 20,900
3	TK-102	1	\$ 94,800
4	TK-103	1	\$ 20,900
5	TK-104	1	\$ 20,300
6	F-101	1	\$ 209,000
7	F-102	1	\$ 209,000
8	Adsorption (T-101)	4	\$ 23,600
9	RO (Modules)	1	\$ 3,700,000
10	Decanter	1	\$ 174,000
11	Diesel Powered Generator	1	\$ 20,000
12	Diesel Engine for Generator	1	\$ 70,000
13	Truck Trailers	10	\$ 400,000
14	P-101 P-104 P-105	3	\$ 6,900
16	P-102 P-103	2	\$ 34,000
17	P-106	1	\$ 23,000
	TOTAL		\$5.13MM

Table 20: Capital Costs

Components of Capital Cost	Details	Cost
Inside Battery Limits (ISBL)	Equipment + Installation	\$5,130,000
Offsite Battery Limit (OSBL)	5% of ISBL	\$256,000
Contingency	10% of ISBL & OSBL	\$538,000
Engineering & Construction	20% of ISBL & OSBL	\$1,030,000
Working Capital	15% of ISBL	\$769,000
Total Capital Cost		\$7.71MM

Table 21: Total Revenue

Description	Quantity (per site) gal	Cost	Total (USD/yr)
Treatment	1,503,671	\$0.15/gal	\$12,400,000
Water	1,444,032	\$0.0083/gal of 30% Volume	\$199,000
Organics	909.15	\$6.2/gal	\$5,640
TOTAL REVENUE			\$12.6 MM

Table 22: Variable Cost of Production (VCOP)

Description	Quantity	Cost	Total (USD/yr)
Raw Materials			
Activated Carbon	8.2 tons/day	\$2000/ton	\$2,710,000
Utilities			
Diesel Fuel for Engine	16.10 gal/hr	\$2.85/gal	\$182,000
Disposal			
Deep-Well Injection	4,197,678 gal	\$0.054/gal	\$197,000
Landfill	18,461.6 tons	\$8.35/ton	\$135,000
Transportation			
3rd Party Trailer Drivers	100 miles, 10 trucks	\$3.39/mile	\$159,0000
TOTAL VCOP			\$3.38MM

Table 23: Fixed Cost of Production (FCOP)

Operations	Description	Cost
Operating Labor	12 Operators \$20.05/hr	\$1,920,000
Supervision	25% Operating Labor	\$481,000
Direct Salary Overhead	45% Operating Labor + Supervision	\$1,080,000
Maintenance	5% ISBL	\$256,000
Property Taxes & Insurances	.5% ISBL + OSBL	\$26,900
Rent of Land	0% ISBL + OSBL	\$0.00
General Plant Overhead	40% Operating Labor + Supervision	\$962,400
Environmental Charges	1% ISBL + OSBL	\$53,800
Royalties	0% Revenue	\$0
Capital Charges	6% Working Capital	\$46,100
Sales & Marketing	5% Total Cost of Production	\$411,000
TOTAL:		\$4.83MM

Table 24: NPV for 5-Year Recovery

Interest=						0.08		
Year	Pre-Tax CF	Depreciation	Taxable Income	Tax	CF	Fd	PV	NPV
0	-\$3.86	0.00	-\$3.86	\$0.00	-\$3.86	1.00	-\$3.86	-\$3.86
1	-\$3.86	0.00	-\$3.86	\$0.00	-\$3.86	0.93	-\$3.57	-\$7.43
2	\$4.40	1.54	\$2.85	\$0.00	\$4.40	0.86	\$3.77	-\$3.66
3	\$4.40	2.47	\$1.93	\$0.57	\$3.82	0.79	\$3.04	-\$0.62
4	\$4.40	1.48	\$2.92	\$0.39	\$4.01	0.74	\$2.95	\$2.32
5	\$4.40	0.89	\$3.51	\$0.58	\$3.81	0.68	\$2.59	\$4.92
6	\$4.40	0.89	\$3.51	\$0.70	\$3.69	0.63	\$2.33	\$7.25
7	\$4.40	0.44	\$3.95	\$0.70	\$3.69	0.58	\$2.16	\$9.40
8	\$4.40	0.00	\$4.40	\$0.79	\$3.61	0.54	\$1.95	\$11.35
9	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.50	\$1.76	\$13.11
10	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.46	\$1.63	\$14.74
11	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.43	\$1.51	\$16.25
12	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.40	\$1.40	\$17.64
13	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.37	\$1.29	\$18.94
14	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.34	\$1.20	\$20.13
15	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.32	\$1.11	\$21.24
16	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.29	\$1.03	\$22.27
17	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.27	\$0.95	\$23.22
18	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.25	\$0.88	\$24.10
19	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.23	\$0.81	\$24.91
20	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.21	\$0.75	\$25.67

Table 25: DCFROR for 5-Year Recovery

						Interest= 0.424246357		
Year	Pre-Tax CF	Depreciation	Taxable Income	Tax	CF	Fd	PV	NPV
0	-\$3.86	0.00	-\$3.86	\$0.00	-\$3.86	1.00	-\$3.86	-\$3.86
1	-\$3.86	0.00	-\$3.86	\$0.00	-\$3.86	0.70	-\$2.71	-\$6.56
2	\$4.40	1.54	\$2.85	\$0.00	\$4.40	0.49	\$2.17	-\$4.40
3	\$4.40	2.47	\$1.93	\$0.57	\$3.82	0.35	\$1.32	-\$3.07
4	\$4.40	1.48	\$2.92	\$0.39	\$4.01	0.24	\$0.97	-\$2.10
5	\$4.40	0.89	\$3.51	\$0.58	\$3.81	0.17	\$0.65	-\$1.45
6	\$4.40	0.89	\$3.51	\$0.70	\$3.69	0.12	\$0.44	-\$1.01
7	\$4.40	0.44	\$3.95	\$0.70	\$3.69	0.08	\$0.31	-\$0.70
8	\$4.40	0.00	\$4.40	\$0.79	\$3.61	0.06	\$0.21	-\$0.48
9	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.04	\$0.15	-\$0.34
10	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.03	\$0.10	-\$0.23
11	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.02	\$0.07	-\$0.16
12	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.01	\$0.05	-\$0.11
13	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.01	\$0.04	-\$0.08
14	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.01	\$0.02	-\$0.05
15	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.00	\$0.02	-\$0.03
16	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.00	\$0.01	-\$0.02
17	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.00	\$0.01	-\$0.01
18	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.00	\$0.01	-\$0.01
19	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.00	\$0.00	\$0.00
20	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.00	\$0.00	\$0.00

Table 26: NPV for 7-Year Recovery

Interest= 0.08								
Year	Pre-Tax CF	Depreciation	Taxable Income	Tax	CF	Fd	PV	NPV
0	-\$3.86	0.00	-\$3.86	\$0.00	-\$3.86	1.00	-\$3.86	-\$3.86
1	-\$3.86	0.00	-\$3.86	\$0.00	-\$3.86	0.93	-\$3.57	-\$7.43
2	\$4.40	1.10	\$3.29	\$0.00	\$4.40	0.86	\$3.77	-\$3.66
3	\$4.40	1.89	\$2.51	\$0.66	\$3.74	0.79	\$2.97	-\$0.69
4	\$4.40	1.35	\$3.05	\$0.50	\$3.89	0.74	\$2.86	\$2.17
5	\$4.40	0.96	\$3.43	\$0.61	\$3.79	0.68	\$2.58	\$4.75
6	\$4.40	0.69	\$3.71	\$0.69	\$3.71	0.63	\$2.34	\$7.08
7	\$4.40	0.69	\$3.71	\$0.74	\$3.65	0.58	\$2.13	\$9.22
8	\$4.40	0.69	\$3.71	\$0.74	\$3.65	0.54	\$1.97	\$11.19
9	\$4.40	0.34	\$4.05	\$0.74	\$3.65	0.50	\$1.83	\$13.02
10	\$4.40	0.00	\$4.40	\$0.81	\$3.59	0.46	\$1.66	\$14.68
11	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.43	\$1.51	\$16.19
12	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.40	\$1.40	\$17.58
13	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.37	\$1.29	\$18.88
14	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.34	\$1.20	\$20.07
15	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.32	\$1.11	\$21.18
16	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.29	\$1.03	\$22.21
17	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.27	\$0.95	\$23.16
18	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.25	\$0.88	\$24.04
19	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.23	\$0.81	\$24.85
20	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.21	\$0.75	\$25.61

Table 27: DCFROR for 7-Year Recovery

						Interest= 0.421131082		
Year	Pre-Tax CF	Depreciation	Taxable Income	Tax	CF	Fd	PV	NPV
0	-\$3.86	0.00	-\$3.86	\$0.00	-\$3.86	1.00	-\$3.86	-\$3.86
1	-\$3.86	0.00	-\$3.86	\$0.00	-\$3.86	0.70	-\$2.71	-\$6.57
2	\$4.40	1.10	\$3.29	\$0.00	\$4.40	0.50	\$2.18	-\$4.39
3	\$4.40	1.89	\$2.51	\$0.66	\$3.74	0.35	\$1.30	-\$3.09
4	\$4.40	1.35	\$3.05	\$0.50	\$3.89	0.25	\$0.95	-\$2.14
5	\$4.40	0.96	\$3.43	\$0.61	\$3.79	0.17	\$0.65	-\$1.48
6	\$4.40	0.69	\$3.71	\$0.69	\$3.71	0.12	\$0.45	-\$1.03
7	\$4.40	0.69	\$3.71	\$0.74	\$3.65	0.09	\$0.31	-\$0.72
8	\$4.40	0.69	\$3.71	\$0.74	\$3.65	0.06	\$0.22	-\$0.50
9	\$4.40	0.34	\$4.05	\$0.74	\$3.65	0.04	\$0.15	-\$0.35
10	\$4.40	0.00	\$4.40	\$0.81	\$3.59	0.03	\$0.11	-\$0.24
11	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.02	\$0.07	-\$0.17
12	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.01	\$0.05	-\$0.12
13	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.01	\$0.04	-\$0.08
14	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.01	\$0.03	-\$0.05
15	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.01	\$0.02	-\$0.04
16	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.00	\$0.01	-\$0.02
17	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.00	\$0.01	-\$0.01
18	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.00	\$0.01	-\$0.01
19	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.00	\$0.00	\$0.00
20	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.00	\$0.00	\$0.00

Table 28: NPV for 15-Year Recovery

							Interest= 0.08	
Year	Pre-Tax CF	Depreciation	Taxable Income	Tax	CF	Fd	PV	NPV
0	-\$3.86	0.00	-\$3.86	\$0.00	-\$3.86	1.00	-\$3.86	-\$3.86
1	-\$3.86	0.00	-\$3.86	\$0.00	-\$3.86	0.93	-\$3.57	-\$7.43
2	\$4.40	0.39	\$4.01	\$0.00	\$4.40	0.86	\$3.77	-\$3.66
3	\$4.40	0.73	\$3.66	\$0.80	\$3.59	0.79	\$2.85	-\$0.81
4	\$4.40	0.66	\$3.74	\$0.73	\$3.66	0.74	\$2.69	\$1.89
5	\$4.40	0.59	\$3.80	\$0.75	\$3.65	0.68	\$2.48	\$4.37
6	\$4.40	0.53	\$3.86	\$0.76	\$3.64	0.63	\$2.29	\$6.66
7	\$4.40	0.48	\$3.92	\$0.77	\$3.62	0.58	\$2.11	\$8.77
8	\$4.40	0.45	\$3.94	\$0.78	\$3.61	0.54	\$1.95	\$10.73
9	\$4.40	0.45	\$3.94	\$0.79	\$3.61	0.50	\$1.80	\$12.53
10	\$4.40	0.46	\$3.94	\$0.79	\$3.61	0.46	\$1.67	\$14.20
11	\$4.40	0.45	\$3.94	\$0.79	\$3.61	0.43	\$1.55	\$15.75
12	\$4.40	0.46	\$3.94	\$0.79	\$3.61	0.40	\$1.43	\$17.18
13	\$4.40	0.45	\$3.94	\$0.79	\$3.61	0.37	\$1.33	\$18.51
14	\$4.40	0.46	\$3.94	\$0.79	\$3.61	0.34	\$1.23	\$19.74
15	\$4.40	0.45	\$3.94	\$0.79	\$3.61	0.32	\$1.14	\$20.87
16	\$4.40	0.46	\$3.94	\$0.79	\$3.61	0.29	\$1.05	\$21.93
17	\$4.40	0.23	\$4.17	\$0.79	\$3.61	0.27	\$0.98	\$22.90
18	\$4.40	0.00	\$4.40	\$0.83	\$3.56	0.25	\$0.89	\$23.79
19	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.23	\$0.81	\$24.61
20	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.21	\$0.75	\$25.36

Table 29: DCFROR for 15-Year Recovery

						Interest= 0.41314691		
Year	Pre-Tax CF	Depreciation	Taxable Income	Tax	CF	Fd	PV	NPV
0	-\$3.86	0.00	-\$3.86	\$0.00	-\$3.86	1.00	-\$3.86	-\$3.86
1	-\$3.86	0.00	-\$3.86	\$0.00	-\$3.86	0.71	-\$2.73	-\$6.59
2	\$4.40	0.39	\$4.01	\$0.00	\$4.40	0.50	\$2.20	-\$4.38
3	\$4.40	0.73	\$3.66	\$0.80	\$3.59	0.35	\$1.27	-\$3.11
4	\$4.40	0.66	\$3.74	\$0.73	\$3.66	0.25	\$0.92	-\$2.19
5	\$4.40	0.59	\$3.80	\$0.75	\$3.65	0.18	\$0.65	-\$1.55
6	\$4.40	0.53	\$3.86	\$0.76	\$3.64	0.13	\$0.46	-\$1.09
7	\$4.40	0.48	\$3.92	\$0.77	\$3.62	0.09	\$0.32	-\$0.77
8	\$4.40	0.45	\$3.94	\$0.78	\$3.61	0.06	\$0.23	-\$0.54
9	\$4.40	0.45	\$3.94	\$0.79	\$3.61	0.04	\$0.16	-\$0.38
10	\$4.40	0.46	\$3.94	\$0.79	\$3.61	0.03	\$0.11	-\$0.27
11	\$4.40	0.45	\$3.94	\$0.79	\$3.61	0.02	\$0.08	-\$0.19
12	\$4.40	0.46	\$3.94	\$0.79	\$3.61	0.02	\$0.06	-\$0.13
13	\$4.40	0.45	\$3.94	\$0.79	\$3.61	0.01	\$0.04	-\$0.09
14	\$4.40	0.46	\$3.94	\$0.79	\$3.61	0.01	\$0.03	-\$0.06
15	\$4.40	0.45	\$3.94	\$0.79	\$3.61	0.01	\$0.02	-\$0.04
16	\$4.40	0.46	\$3.94	\$0.79	\$3.61	0.00	\$0.01	-\$0.03
17	\$4.40	0.23	\$4.17	\$0.79	\$3.61	0.00	\$0.01	-\$0.02
18	\$4.40	0.00	\$4.40	\$0.83	\$3.56	0.00	\$0.01	-\$0.01
19	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.00	\$0.00	\$0.00
20	\$4.40	0.00	\$4.40	\$0.88	\$3.52	0.00	\$0.00	\$0.00

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